International Journal of Hydrogen Energy 178 (2025) 151474

Contents lists available at ScienceDirect

International Journal of Hydrogen Energy

journal homepage: www.elsevier.com/locate/he

Techno-economic analysis of a hydrogen-based hybrid renewable energy system for off-grid power supply in Kenya's urban area, considering the impact of the hydrogen market participation on the economic viability of the system

Alphonce Ngila Mulumba [a], Hooman Farzaneh [a,b,c,*]

[a] Interdisciplinary Graduate School of Engineering Sciences, Kyushu University, Fukuoka, 816-8580, Japan
[b] Transdisciplinary Research and Education Center for Green Technologies, Kyushu University, Fukuoka, Japan
[c] Faculty of Engineering Sciences, Kyushu University, Fukuoka, Japan

A R T I C L E I N F O

Handling editor: Prof. A.B. Basile

Keywords:
Techno-economic assessment
Hybrid renewable energy system
Hydrogen market
Multi-objective optimization

A B S T R A C T

This study develops a techno-economic framework to optimize a hydrogen-based hybrid renewable energy system (HRES) combining photovoltaic, wind, biomass, and fuel cell technologies for off-grid supply in Embakasi Pipeline estate, Nairobi. The framework applies an ε-constraint-based multi-objective mixed-integer linear programming model that optimizes system configuration for both cost-effectiveness and reliability. A key innovation of this study is addressing three critical research gaps: (i) integrating local hydrogen market participation into operational optimization through surplus hydrogen trading, (ii) providing a comprehensive techno-economic feasibility assessment of hydrogen-based HRES in Kenya, and (iii) incorporating a unit commitment strategy through dynamic dispatch scheduling. Results indicate that hydrogen market participation reduces the cost of electricity (COE) by \$0.174/kWh to \$0.494/kWh, while achieving a reliability index of 0.961, compared to a 0.978 index at \$0.668/kWh without market trading. Sensitivity analysis shows that higher hydrogen prices (USD 9.9/kg) can further lower COE from \$0.494/kWh to \$0.405/kWh.

1. Introduction

1.1. Background and motivation

The global shift towards renewable energy is an unprecedented trend of the twenty-first century, driven by the urgent need for sustainable energy solutions. The G20 recently highlighted the importance of incorporating renewable energy into national electricity grids to address growing energy needs and combat climate change. Africa, and mainly Sub-Saharan Africa, including Kenya, actively participate in this global initiative, with renewable energy sources making significant inroads into the electricity market [1]. Africa is increasingly gaining global attention as a potential renewable energy powerhouse capable of leading the decarbonization of energy systems and industrial processes. Scaling up renewable energy development across the continent is critical not only for driving inclusive economic growth and diversifying electricity supply, but also for reducing dependency on imported fossil fuels and mitigating greenhouse gas emissions in a carbon-constrained global environment. Kenya's commitment to renewable energy integration is reinforced by the Vision 2030 policy, which outlines an ambitious goal of achieving 100 % access to clean energy by 2050 [2]. As part of this initiative, the Renewable Energy Integration (REI) Program aims to assist Kenya in achieving its Nationally Determined Contributions (NDC) target of reducing greenhouse gas emissions by 32 % by 2030 [3]. Currently, Kenya's energy matrix comprises 80 % petroleum-based fossil fuels, 18 % renewable energy, and 2 % coal. Notably, over 85 % of the population relies on wood biomass, with 86 % in rural areas and 21 % in urban regions [4].

The growing adoption of renewable energy systems, including the hybridization of various power sources with storage technologies, holds significant promise for Kenya. Hybrid renewable energy systems (HRES), particularly those optimized for site-specific conditions, offer promising pathways for clean energy deployment [5]. However, they often face systemic challenges such as intermittency management,

* Corresponding author. Interdisciplinary Graduate School of Engineering Sciences, Kyushu University, Fukuoka, 816-8580, Japan.
E-mail address: farzaneh.hooman.961@m.kyushu-u.ac.jp (H. Farzaneh).

https://doi.org/10.1016/j.ijhydene.2025.151474
Received 6 May 2025; Received in revised form 8 September 2025; Accepted 9 September 2025
Available online 23 September 2025

integration complexity, grid instability, and capital cost uncertainties [6]. One innovative solution involves leveraging surplus electricity generated during peak solar or wind periods for green hydrogen production. This not only supports secondary power generation via fuel cells but also enables hydrogen commercialization, reducing overall electricity costs and enhancing energy system flexibility through price adjustment mechanisms embedded within generation strategies. The integration of hydrogen storage with the HRES not only enhances system reliability during renewable intermittency but also enables grid independence and additional revenue streams. Such a framework paves the way for economically viable green hydrogen-based HRES under evolving market and regulatory conditions [7]. Moreover, the economic viability of hydrogen-based HRES is affected by the availability of renewable resources and the pricing mechanisms for both electricity and hydrogen. This necessitates favorable policy frameworks and the design of cost-effective systems [8]. These complexities further challenge the integration of hydrogen into HRES configurations and underscore the need for comprehensive techno-economic assessment studies.

Kenya's abundant solar and wind resources make it well-suited for large-scale hydrogen production via electrolysis [9]. Moreover, hydrogen's high energy density allows it to store more energy in a smaller volume compared to conventional battery systems, offering a distinct advantage in a country where spatial constraints can limit infrastructure deployment [10]. Additionally, the decarbonization potential of hydrogen aligns with Kenya's commitment to sustainable development and climate action, aiding in the reduction of greenhouse gas emissions while meeting energy needs [11]. The low degradation rate and longevity of hydrogen storage systems enhance the viability of infrastructure investments, offering a durable and future-proof energy solution [12]. With these attributes, hydrogen emerges as a compelling alternative for strengthening Kenya's energy system resilience and sustainability in the near future. Furthermore, trading surplus hydrogen produced by the HRES in the local markets provides a dual advantage by utilizing Kenya's abundant renewable energy resources and promoting energy diversification and economic growth [13].

1.2. *Literature review*

1.2.1. *Hydrogen based-HRES utilization*

Recent studies have investigated the techno-economic viability of hydrogen-based HRES, highlighting hydrogen's potential as a clean energy carrier across various locations around the world [14]. Though hydrogen-based systems typically involve higher upfront costs than batteries, they provide critical advantages for hybrid renewable energy setups. Hydrogen enhances energy security by enabling long-duration storage from hours to seasons, offering greater resilience than batteries in managing intermittent solar and wind output. As highlighted in Ref. [15], systems using electrolyzers, hydrogen tanks, and fuel cells ensure continuous off-grid power, especially in fluctuating weather patterns. When powered by renewable sources, green hydrogen reduces fossil fuel reliance and supports decarbonization in sectors like transport and industry. Long-term benefits include lower emissions, extended storage lifespan, and improved system flexibility, making hydrogen a strategic complement rather than a substitute to batteries in meeting the low-carbon energy targets. Urs et al. (2023) analyzed a grid-connected PV-reversible fuel cell (RFC) microgrid for an office building in the UAE, demonstrating that green hydrogen production could enhance grid independence. Their study exhibited green hydrogen production from excess solar energy, storage, and utilization in fuel cell generators to meet demand during non-generation hours, enhancing self-sufficiency and reducing grid dependence. Further, demonstrated hydrogen-based energy storage satisfies 16.7 % of load demand, contributing to energy reliability and price stability [16]. Similarly, a comprehensive techno-economic simulation by Kharel and Shabani (2018) explored multiple hybrid renewable energy configurations in South Australia, using HOMER software. Notably, a system combining PV, wind, batteries, and hydrogen storage, integrating an electrolyzer, hydrogen tank, and fuel cell, achieved a significantly lower cost of electricity (COE) at $0.626/kWh. In comparison, the PV/wind/battery-only configuration yielded a much higher COE of $2.68/kWh, mainly due to the costly requirement for oversized and frequently replaced battery banks. The hydrogen-based system's cost advantage stemmed from its ability to reduce battery reliance, provide long-duration storage, and improve energy dispatch efficiency. Moreover, monetizing surplus green hydrogen further reduced the COE to $0.494/kWh, underscoring hydrogen's dual role in storage and revenue generation [17].

In the case of Sub-Saharan Africa, Adedoja et al. evaluated the techno-economic evaluation of hydrogen-based HRES for powering rural healthcare facilities in Nigeria and South Africa, developing an optimization mode based on the Evaluation based on Distance from Average Solution (EDAS) to optimize the configuration of the system, combining solar PV, wind, batteries, fuel cells, electrolyzers, and hydrogen storage. The results indicated that the levelized cost of energy (LCOE) ranged from $0.410 to $0.633 per kWh. Photovoltaic systems were responsible for supplying over 90 % of the energy load, showcasing reliable power delivery and significant emission reductions of up to 100 %. Their findings emphasized the importance of context-specific design for enhancing energy access and healthcare services in Sub-Saharan Africa's remote regions [18]. Hydrogen production via PV/Wind/Electrolysis in Sub-Saharan Africa, particularly in Cameroon, is found to be more cost-effective than Wind/Electrolysis alone due to the region's higher solar potential. PV systems meet most load requirements, reducing reliance on expensive wind infrastructure. The hybrid configuration promotes better resource utilization, optimizing electrolyzer efficiency and lowering overall system costs across various locations [19]. While the techno-economic comparison of HRES in Sub-Saharan Africa shows that PV/Battery systems have the lowest LCOE at 0.255 €/kWh, integrating hydrogen through fuel cells enhances the system's flexibility and reliability [20]. Ayodele et al. investigated the integration of hydrogen in a hybrid renewable energy microgrid designed to supply power to a rural health clinic in South Africa. The system comprising PV modules, wind turbines, an electrolyzer, hydrogen storage, and a fuel cell demonstrated reliable off-grid performance. Although the levelized cost of energy (LCOE) was relatively high at $2.34/kWh, the study established its viability by comparing it with the cost of grid extension. Specifically, the proposed system had a breakeven grid extension distance of 8.81 km, while the actual distance to the nearest grid point was 21.35 km, making the off-grid HRES a more cost-effective and practical solution for the remote clinic [21]. However, limited studies on the utilization of hydrogen-based HRES cases in Africa highlight a significant lack of policy support, infrastructure, and deployment strategies. This situation emphasizes both the feasibility and the untapped potential of hydrogen-based HRES to deliver sustainable and resilient energy access throughout Sub-Saharan Africa.

1.2.2. *Modeling approaches used in the optimal design of hydrogen-based based-HRES*

As previously explained, earlier studies have shown that integrating hydrogen can significantly affect the cost of energy (COE) of HRES [22]. However, the economic feasibility of these systems is heavily affected by the dynamic and unpredictable weather conditions in the specific area being studied. This highlights the importance of developing robust dynamic optimization methods for evaluating the techno-economic feasibility of a hydrogen-based HRES. The conceptual application of dynamic optimization in HRES facilitates the strategic allocation of renewable energy resources based on real-time demand patterns and associated cost implications [23]. In essence, it involves continuously adapting the operation of the system to maximize performance based on predefined objectives like minimizing cost, maximizing reliability, or mitigating emissions [24]. Most previous studies utilized a cost-effectiveness approach that minimizes total system costs, including operational, capital, and transmission expenses, while also considering CO_2-equivalent emissions through environmental taxes [25,26]. For instance, Tatar et al. utilized a mixed-integer linear programming (MILP) approach for optimizing a hydrogen-based HRES, considering green hydrogen production and synthetic methanation under weather and demand uncertainty. Their study evaluated multiple scenarios, identifying PV and bio-gasifiers as cost-effective, low-emission solutions, incorporating real-time decision-making for green hydrogen storage and commercialization [27]. An optimization modeling framework that utilizes the particle swarm optimization (PSO) method for the optimal design of a hydrogen-based HRES is developed in Ref. [28]. In their proposed HRES, hydrogen is produced using both solar electrolysis and supercritical water gasification (SCWG) of biomass feedstock. The study explored two scenarios: one aimed at minimizing the total system cost in an off-grid mode, and the other focused on maximizing profit under a grid-tied feed-in tariff (FiT) scheme. The findings indicate that the proposed hydrogen-based HRES can successfully meet the load requirements for residential areas in Fukushima prefecture in Japan, while maintaining a reasonable levelized cost of energy (LCOE).

Finding an optimal configuration of a hydrogen-based HRES that balances cost and reliability requires considering technical, economic, and environmental factors within the dispatch or scheduling framework, often resulting in trade-offs between conflicting objectives. One important factor to consider is the order of power dispatch, also referred to as the generation scheduling framework. This framework consists of two primary components: unit commitment and economic dispatch. Unit commitment determines the optimal schedule for operating power generators to meet energy demand, while economic dispatch identifies the most cost-effective operating levels, minimizing operational costs and ensuring stable system operation [29]. The dynamic dispatch scheduling enables the system to adapt in real time to changes in renewable energy generation and load demand. This capability is crucial for effectively managing hydrogen production, storage, and use in the hydrogen-based HRES, especially considering the variability associated with solar and wind energy sources. Previous studies have developed numerous techniques to investigate optimal **unit commitment** and **economic dispatch** in HRES. For instance, a recently proposed two-stage risk-constrained stochastic programming model for hydrogen-based rural microgrids seeks to optimize investment and long-term economic dispatch under uncertainty through scenario sampling and decomposition [30]. While it demonstrated efficiency in handling multi-energy integration and resilience, it relies heavily on sampling-based risk control and complex decomposition, which may limit its adaptability to dynamic operational changes. In another effort, Moghaddam et al. employ the Flower Pollination Algorithm (FPA) to optimize the design and energy management of a PV/Wind/Fuel Cell hybrid system with hydrogen storage, focusing on minimizing total net present cost (TNPC) under reliability constraints. Although FPA achieves fast convergence and outperforms PSO in certain aspects, its heuristic nature lacks guaranteed global optimality and real-time adaptability. Alternatively, the integration of the ε-constraint multi-objective MILP framework with dynamic energy dispatch scheduling offers more granular control over competing objectives and supports real-time responsiveness, making it more suitable for adaptive, real-world hydrogen-based HRES design in uncertain environments [31]. In contrast to the optimization methods used in the above-mentioned studies, the ε-constraint MILP framework enables deterministic, structured optimization with dynamic dispatch, offering enhanced control over trade-offs such as cost and reliability [32].

Table 1 summarizes recent studies on hydrogen-based HRES, highlighting system configurations and the optimization modeling strategies employed.

Table 1
Previous studies on Hydrogen-based HRES optimization modeling.

Ref.	System Configuration						Technique	Objective & Contribution	Findings & Research Gap
	PV	Wind	Bm	BESS	FC	HESS			
[33]	✓	✓	×	×	✓	✓	PSO	Develops a cost-effective design and energy management strategy for optimizing DERs, focusing on H_2 storage	Demonstrates cost reduction but lacks integration of real-time hydrogen market dynamics and advanced dispatch coordination.
[30]	✓	✓	×	×	✓	✓	MILP	Models an H_2-based rural microgrid optimizing multi-energy resources to minimize LCOE	Achieves cost savings but limited operational adaptability; no link to hydrogen sales or seasonal market participation.
[34]	✓	✓	×	×	✓	✓	HOMER	Evaluates H_2-HRES under multi-criteria cost and viability analysis	Identifies optimal tech mix but relies on static economic metrics, overlooking hourly scheduling and market-driven operations.
[32]	✓	✓	×	×	✓	✓	Flower Pollination Algorithm (FPA)	Optimizes cost and reliability in H_2-HRES using metaheuristics	Offers improved reliability but oversimplifies variability in demand, weather, and market prices, limiting real-world applicability.
[35]	✓	✓	✓	✓	✓	✓	HOMER, GA	Sizes H_2-HRES for remote areas, minimizing COE and CO_2 emissions	Reduces emissions but is sensitive to input fluctuations; no dynamic link between hydrogen storage, dispatch, and market sales.
[36]	✓	✓	×	✓	✓	✓	BMILP	Presents a bi-level planning model for energy hub-integrated systems, optimizing Hydrogen production and minimizing total cost (TAC) and levelized cost of hydrogen (LCOH).	Handles multi-objective trade-offs but is computationally heavy; lacks explicit integration of hydrogen market revenue streams.
[37]	✓	✓	✓	✓	✓	✓	P-Graph	Island energy model optimizing H_2 and battery storage to minimize COE and CO2 emissions, demonstrating cost savings.	Shows cost savings but lacks operational flexibility to adapt to fluctuating renewable supply and market demand.
[38]	✓	×	×	✓	✓	✓	HOMER	Optimizes PV-based green hydrogen production and minimizes levelized hydrogen cost (LHC) for cost-effectiveness.	Achieves low LHC but is highly sensitive to initial assumptions; no adaptive market-linked dispatch mechanism.
This study	✓	✓	✓	×	✓	✓	ε-constraint MILP	Integrates dynamic local hydrogen market participation with unit commitment to minimize COE and maximize reliability.	Demonstrates that surplus hydrogen trading, coupled with dynamic dispatch and seasonal adaptability, can reduce COE while maintaining operational resilience in the sub-Saharan Africa urban context.

1.3. Research contributions

Based on the above discussion, three critical research gaps are evident.

1) Limited integration of hydrogen market participation in operational optimization.

Existing research on hydrogen-based HRES seldom incorporates local hydrogen market participation into operational optimization frameworks. This study addresses that gap by explicitly modeling surplus hydrogen trading in local energy markets and integrating it into a dynamic configuration of HRES, an approach with high relevance to African urban contexts.

2) Lack of comprehensive techno-economic feasibility studies for hydrogen-based HRES in Kenya. Despite Kenya's abundant solar resources and strong potential for large-scale green hydrogen production, there is a scarcity of comprehensive techno-economic assessments for hydrogen-based HRES targeting urban power supply. Few studies evaluate the decarbonization potential of green hydrogen in advancing national sustainability targets.
3) Overlooking unit commitment in optimal scheduling. Most previous studies focus on cost-effectiveness in HRES design but neglect the role of unit commitment in scheduling renewable generation for efficient hydrogen production, storage, and utilization factors, which strongly affect the economic viability of the system.

To address the above gaps, this study develops a novel techno-economic assessment modeling framework in order to find the optimal configuration of a proposed hydrogen-based HRES for off-grid urban power supply in Embakasi Pipeline estate, Nairobi, Kenya. The proposed HRES consists of solar PV, wind turbines, a fuel cell, and a biomass generator, with a solar-powered electrolyzer, ensuring hydrogen production during surplus electricity generation. To further promote the commercialization of green hydrogen, the surplus hydrogen produced by the system is intended to be traded in a local hydrogen market in the study area to increase the economic feasibility of the scheme. Unlike conventional models, the modeling approach developed in this study uniquely incorporates hydrogen market participation dynamics within a multi-objective optimization framework, which optimizes the configuration of the system, considering both cost-effectiveness and system's reliability. A key innovation in the developed modeling approach is the inclusion of the unit commitment strategy through dynamic dispatch scheduling, which enhances hydrogen market participation by facilitating optimal power dispatch between renewable energy supply and load demand. This formulation explicitly considers market demand for hydrogen sales, dynamically adjusting the system's reliance on power generation from the fuel cell and trading surplus hydrogen in the market. The model systematically balances trade-offs between the cost of energy (COE) and system reliability, using a ε-constraint multi-objective MILP framework, demonstrating that strategic hydrogen sales can reduce COE, while maintaining operational resilience. Seasonal adaptability further enhances the model: during dry seasons, abundant renewable output supports increased hydrogen sales, whereas wet seasons with reduced generation shift reliance toward fuel cell utilization. This seasonal flexibility directly impacts both COE and hydrogen reserve levels. Additionally, this study incorporates a GIS-based biomass resource assessment to optimize local feedstock integration, improving system reliability. The higher heating value (HHV) of biomass is estimated using a Gibbs free energy minimization sub-model, which introduces a thermodynamically consistent approach to biomass energy potential estimation in the absence of complete empirical data. By linking hydrogen market participation with system optimization, this study provides a scalable, cost-effective framework for off-grid hydrogen-based HRES solutions, demonstrating a pathway to integrate hydrogen economics into power systems for sustainable urban electrification.

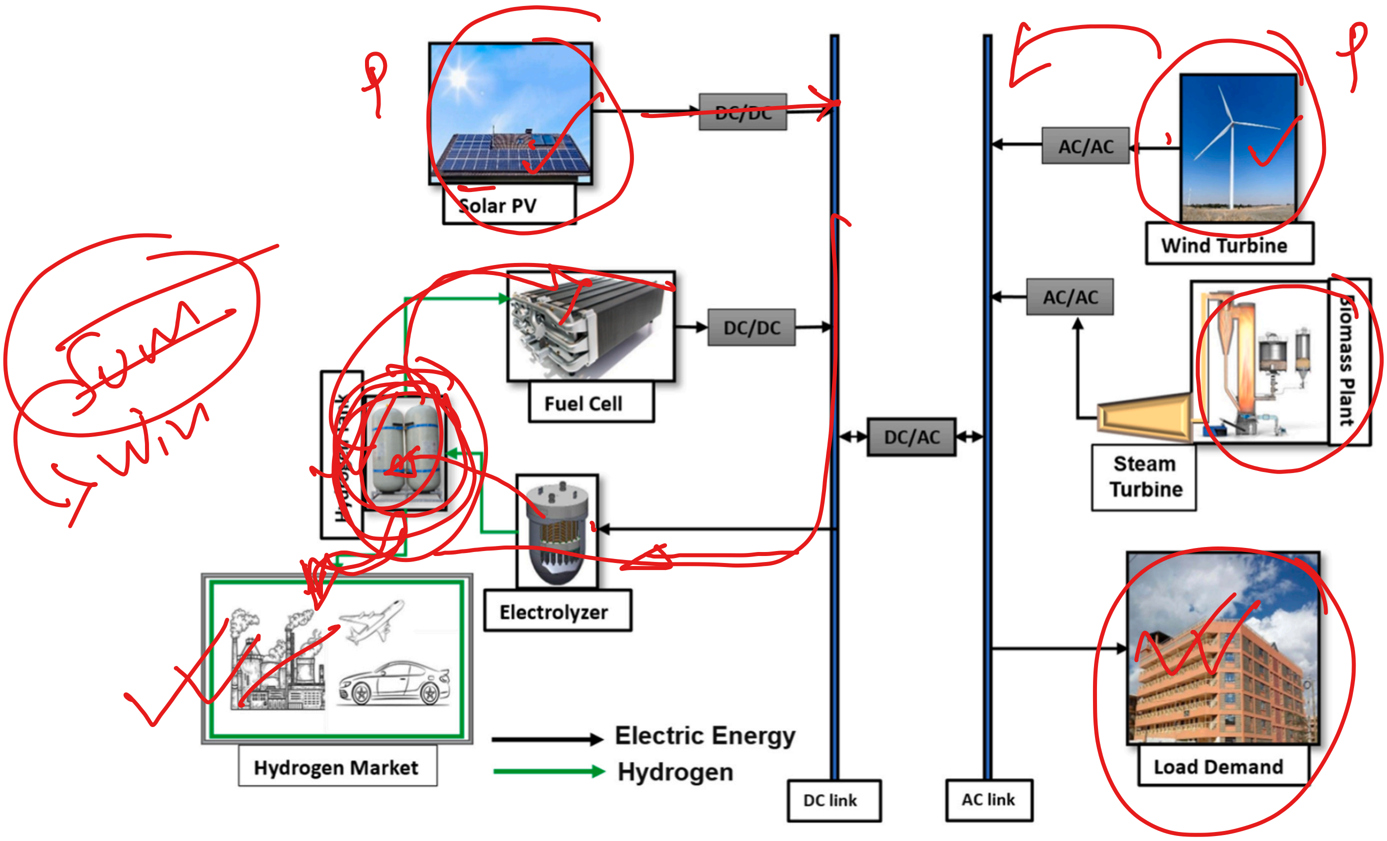


Fig. 1. Schematic diagram of the Proposed H2-HRES in this study.

The rest of this paper is organized as follows: Section 2 outlines the system configuration and delves into the optimization of the hydrogen-based HRES system sizing, detailing the proposed multi-objective optimization approach alongside the mathematical modeling of power system components. Section 3 presents the results and discussions, including the verification of the optimization results through comparison with existing work. Lastly, sections 4 and 5 present this study's limitations, concluding remarks, and future research work, respectively.

2. Methodology

2.1. System configuration

The proposed hydrogen-based HRES (H_2-HRES) analyzed in this study integrates multiple technologies to create a sustainable and resilient off-grid microgrid. The system consists of a solar PV, wind turbine generator, biomass generator, proton exchange membrane (PEM) fuel cell, and solar water electrolyzer, working together to meet the annual electricity demand for a selected residential area in Nairobi city. Fig. 1 illustrates the system configuration, showing how power generation, energy storage, and electrical load demand are interconnected.

The primary power generators are the PV and wind turbines. Solar PV convert sunlight into direct current (DC) electricity, which is either supplied directly to the load or used to produce hydrogen in a PEM water electrolyzer, when surplus energy is available. The PEM electrolyzer splits water molecules into hydrogen and oxygen, with the hydrogen stored in high-pressure tanks for later use. Similarly, the wind turbine generates alternating current (AC) electricity, which is prioritized for load consumption. However, since solar and wind power are intermittent, additional backup power sources are required. When the electricity generated by these sources is insufficient, either the biomass-powered Rankine cycle generator or the PEM fuel cell is activated. The biomass generator utilizes feedstock to produce steam that drives a turbine, generating electricity, while the PEM fuel cell converts hydrogen produced by the electrolyzer into electricity. During periods of high electricity demand or low renewable generation, the stored hydrogen is fed into the PEM fuel cell to generate electricity, ensuring a continuous power supply. This system design enhances energy reliability by effectively balancing renewable generation, energy storage, and hydrogen utilization. The integration of multiple power sources and energy storage solutions ensures that the residential load demand is consistently met while optimizing cost and resource efficiency, making the H_2-HRES a viable and sustainable solution for off-grid energy generation.

2.2. Study region – Embakasi Pipeline Estate (Nairobi, Kenya)

Embakasi Pipeline Estate is a densely populated urban neighborhood in eastern Nairobi, predominantly inhabited by low- and middle-income households. As of the 2019 census, the area recorded a population of 166,517 residents, with an exceptionally high density of 106,445 persons per square kilometer [39]. The estate has expanded rapidly due to rural-urban migration and urban sprawl, resulting in a mix of formal and informal housing and correspondingly high energy demand density [40]. Despite its proximity to Nairobi's main transmission corridors, Pipeline suffers from unreliable electricity supply, frequent outages, and dependence on decentralized diesel generators. Over 46 % of households turn to charcoal as a fallback cooking fuel, reflecting its role as a substitute during periods of electricity shortage or high cost [41]. This reliance reinforces both indoor and outdoor air pollution, contributing to significant health and environmental burdens. Using Kenya's average per-capita electricity consumption of 191 kWh per year, the 2019 population corresponds to a baseline annual demand of about 31.8 GWh [42]. With the population projected to grow by approximately 34 % between 2020 and 2045, demand is expected to reach 42.6 GWh per year, even under conservative assumptions of constant per-capita use.

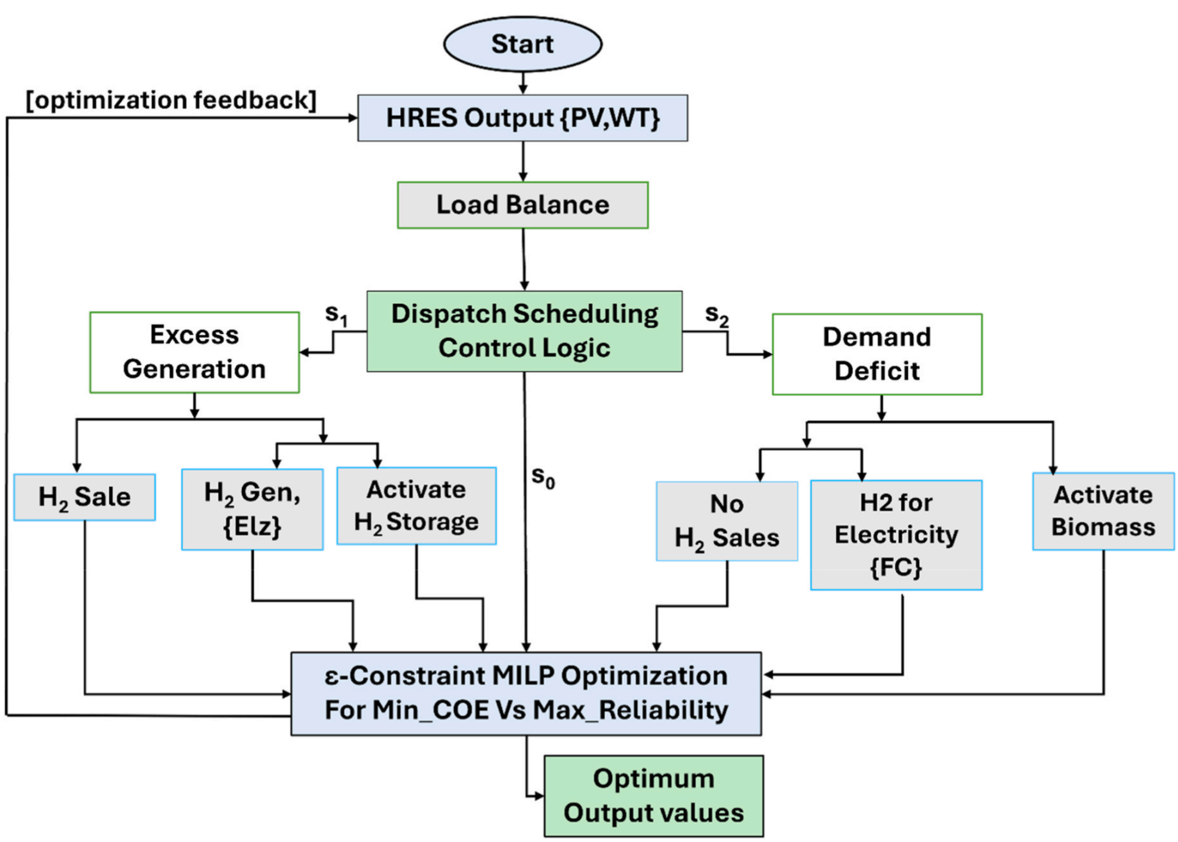


Fig. 2. Proposed operation logic flow diagram.

Given Pipeline's extreme density, this additional demand will be highly concentrated, further straining fragile local infrastructure [43]. Such concentrated urban growth provides a natural entry point for renewable hydrogen, which can serve both as a balancing resource for electricity supply and as a commercial energy carrier for Nairobi's emerging sectors.

2.3. Local hydrogen market structure and assumptions

In this study, the local hydrogen market is modeled as a competitive, exogenous-price market integrated into the operational optimization of the hydrogen-based HRES. Hydrogen is assumed to be sold exclusively to local industrial and agricultural users within the Nairobi metropolitan area, with a priority focus on fertilizer production in line with Kenya's Green Hydrogen Strategy, which emphasizes building domestic consumption before considering exports [44]. Market prices are treated as exogenous inputs, derived from recent techno-economic estimates for green hydrogen production costs in Kenya (USD 2.7–9.9/kgH_2) and are considered to vary seasonally to reflect changes in production costs and demand patterns [45]. The available hydrogen storage constrains sales volumes after maintaining a minimum operational reserve and by daily transaction capacity limits, representing current distribution infrastructure [46]. The assumed market price includes compression, transportation, and handling costs, avoiding the need to model these separately [47]. From a regulatory perspective, the hydrogen market in Kenya is at an early stage of development [48], currently overseen by the Green Hydrogen Secretariat and Working Group under existing commercial frameworks, with no international trade component. These structural and economic assumptions are embedded in the objective function (Eq. (2)) through the hydrogen revenue term and in market participation constraints that regulate trading decisions based on demand, price signals, and storage status.

It is noted that, Kenya's National Green Hydrogen Strategy (2023–2024) aims to produce between 300,000 and 400,000 tonnes of green fertilizer annually by 2032. This goal will be supported by an electrolyzer capacity of 150–250 MW, an investment of USD 1 billion, and the creation of 25,000 jobs. With hydrogen demand expected to grow across agriculture, industry, power, and transport sectors, a strong market framework is essential. This framework should accommodate dynamic pricing, facilitate infrastructure expansion, and support various applications to ensure adaptability to future market conditions.

2.4. Proposed operation logic

The proposed hydrogen-based HRES employs a dynamic dispatch strategy that coordinates renewable generation, hydrogen production, storage, utilization, and market participation to ensure cost-effective and reliable power supply, as detailed in Fig. 2. Electricity from solar PV and wind turbines is first allocated to meet the instantaneous load (situation, S_0), as enforced by the supply–demand balance constraint (Eq. (8)). When generation exceeds demand (situation S_1), surplus electricity is directed to the PEM electrolyzer, which operates within its rated capacity and minimum load limits, producing hydrogen subject to capacity constraints. Stored hydrogen is monitored hourly to maintain operational reserves and assess eligibility for sale.

If renewable generation is insufficient (situation S_2), the system dispatches either the biomass generator or the PEM fuel cell, depending on hydrogen availability, generator capacity, and cost-minimization criteria, as governed by unit commitment constraints displayed in Fig. 4. Hydrogen is sold in the local market, with traded quantities determined by market price signals, storage levels, and demand satisfaction, through market participation constraints. Seasonal adaptability is embedded: high renewable output in dry seasons increases hydrogen sales, while reduced generation in wet seasons shifts reliance toward fuel cells and biomass to prioritize reliability.

2.5. Model development

In this study, the developed multi-objective optimization model includes two competing objective functions to achieve the best trade-off between the system's cost of electricity (COE) and the ultimate reliability indicator, signified by the unmet energy (UME) parameter. The optimal system sizing is established by accounting for the costs of all components, installation, and operation. The desired power reliability is achieved by maximizing generation to meet demand while minimizing the cost of electricity to the greatest extent possible.

The mathematical representation for the multi-objective optimization model, which integrates both cost of electricity and energy reliability objective functions, using the ε-constraint method, is formulated as follows [31]:

$$\min F(x) = \begin{pmatrix} f_1(COE) \\ f_2(UME) \end{pmatrix} \rightarrow \begin{cases} \min f_1(COE) \\ \min f_2(UME) \\ \epsilon_{min} \le \epsilon \le \epsilon_{max} \end{cases} \tag{1}$$

The ε-constraint method used in the developed multi-objective optimization model in this study implies minimizing one objective function, while converting the other into a constraint with an upper bound epsilon(ε). This approach streamlines the problem by focusing on optimizing one goal: $f_1(COE)$ and controlling the other, $f_2(UME)$, within acceptable limits. By varying the epsilon(ε), exploration of numerous trade-offs between the objectives is obtained. This method provides an accurate way to balance conflicting goals, making it easier to analyze how changes in the constrained objective affect the primary objective [31]. The flow chart in Fig. 3 illustrates the integration of the ε-constraint logic into the multi-objective optimization system.

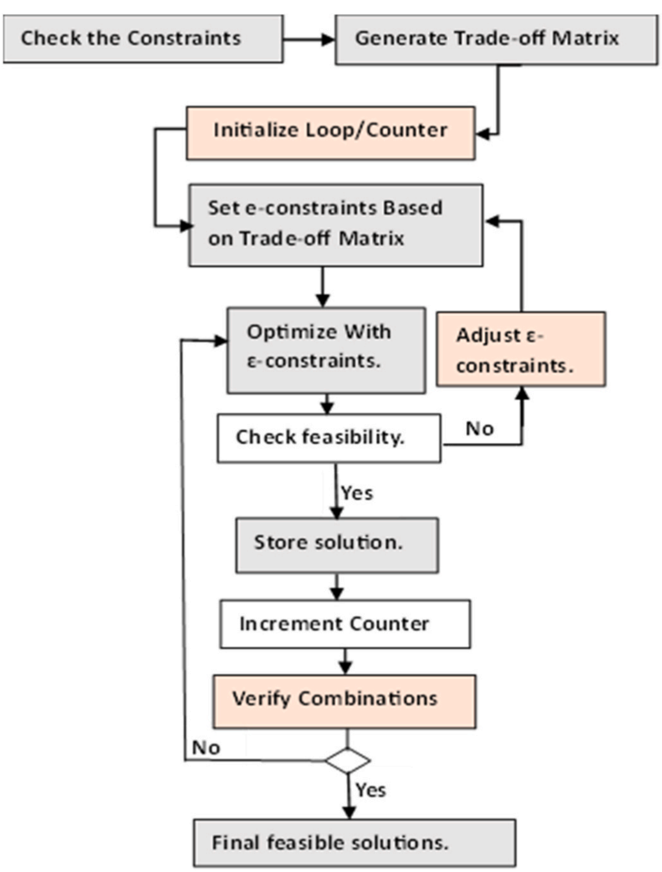
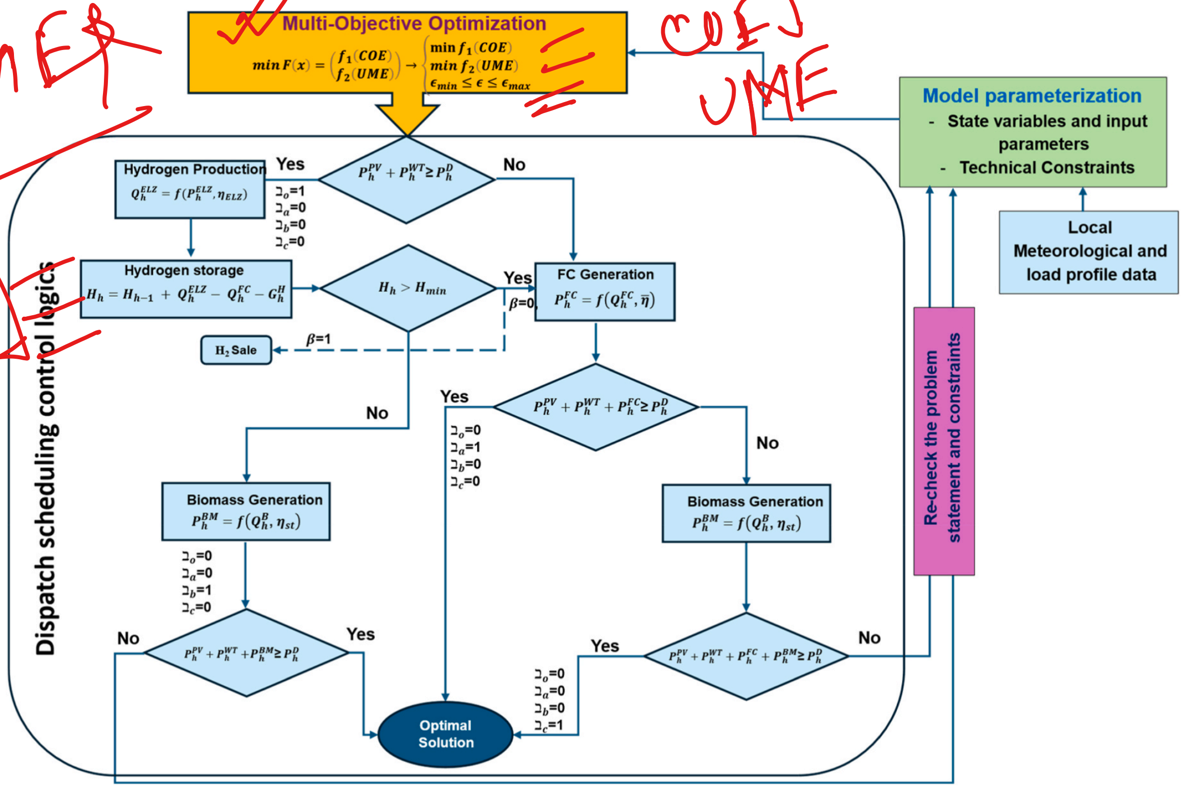


Fig. 3. ε-constraint logic used in the developed multi-objective optimization model in this study.

Loop that forces the system to test different reliability levels one by one

HOMER

COE UME



COE

Fig. 4. Overall structure of the dispatch model developed in this study with detailed unit commitment strategy.

2.5.1. *Objective functions*

The total capital expense, along with the annual operation, maintenance, and hydrogen storage costs, is aggregated to evaluate the COE for the proposed H_2-HRES according to equation (2). The revenue generated from the annual hydrogen available for sale (G^H), at an average market price of p^H curtailments on the COE. β is a binary activation variable that is assigned one (1) state whenever excess hydrogen is available for sale and a zero (0) state in the contrary situation [31]. The total cost, TC including unit cost (λ) of capital expense along with the annual operation, maintenance, and hydrogen storage costs for every type of generator τ, are represented by the expression (3) and further detailed in (4).

$$f_1(COE) = \left(\left[\frac{TC - [\beta \cdot (G^H \cdot p^H)]}{(1+dr)^n} \right] \Big/ \left[\frac{P^{HRES}}{(1+dr)^n} \right] \right) \tag{2}$$

$$TC = \sum_{h=1}^{8760} \sum_{\tau} P_{h\tau}^{HRES} \lambda_\tau \tag{3}$$

$$\lambda_\tau = \lambda_{PV} + \lambda_{WT} + \lambda_{ELZ} + \lambda_{FC} + \lambda_{H_{st}} + \lambda_{OM} \tag{4}$$

Power generation is optimally produced to satisfy electrical load requirements by hybridizing PV modules and wind turbines. When solar and wind power generations are not available, a backup generation is provided, using biomass and fuel cell systems. The total power generated by the HRES system P^{HRES}, is expressed as follows:

$$P^{HRES} = \sum_{h=1}^{8760} (P_h^{PV} + P_h^{WT} + P_h^{BM} + P_h^{FC}) \tag{5}$$

$f_2(UME)$ is formulated based on the value of the reliability index (Ri), ranging from 0 to 1 such that 1 represents perfect reliability and 0 indicates complete system failure [31]:

$$Ri_h = 1 - UME_h \tag{6}$$

$$f_2(UME) = \sum_{h=1}^{8760} \left\{ \frac{((P_h^D + P_h^{ELZ}) - P_h^{HRES})}{P_h^{HRES}} \right\} \tag{7}$$

2.5.2. *Technical constraints*

2.5.2.1. Dynamic dispatch scheduling. A power dynamic dispatch control strategy ensures the supply-demand balance equation (8), which combines output from generators and sustains both the electrical load and the power requirements of the electrolyzer.

$$P_h^{HRES} = P_h^D + P_h^{ELZ} \tag{8}$$

The dispatch logic is designed to handle the fluctuating nature of renewable energy sources by leveraging biomass and fuel cell generators as backup power sources [49]. This is modeled by allocating four load balance modes to the envisaged operational mode and assigning binary variables as switching flags $(\beth_0, \beth_a, \beth_b, \beth_c)$ to each mode equivalence as demonstrated in equation (9). The complementary generators are enabled to generate with respect to this control strategy if and only if assigned state one (1); otherwise, the output remains zero when the allocated binary state is zero (0). For example, when there is ample solar and/or wind power during the day, excess electricity is directed to the electrolyzer by activating the binary logic $\beth_0$ permitting the generation of hydrogen. Trigger states $\beth_a$, $\beth_b$ and $\beth_c$ remain at zero, turning off biomass and fuel cell generators during this phase. This flexible, dynamic dispatch scheduling approach, illustrated in Fig. 3, ensures that the system maintains continuous, reliable electricity generation, even during periods of low renewable energy availability [50]:

$$P_h^{HRES} = \{ \beth_o \cdot (P_h^{PV} + P_h^{WT}), \beth_a \cdot (P_h^{PV} + P_h^{WT} + P_h^{FC}), \beth_b \cdot (P_h^{PV} + P_h^{WT} + P_h^{BM}), \beth_c \cdot (P_h^{PV} + P_h^{WT} + P_h^{BM} + P_h^{FC}) \} \tag{9}$$

And

$$\beth_o, \beth_a, \beth_b, \beth_c \in \{0,1\} \tag{10}$$

2.5.2.2. PV and wind turbine power generation. The solar PV power output is formulated as follows [51]:

$$P_h^{PV} = (PV_{max} N_p Pv_{df}) \frac{G_h}{G_{STC}} \left(1 + K_T \left(T_c - \left(T_{ah} + \frac{G_h}{800}(T_{NOCT} - 20)\right)\right)\right) \tag{11}$$

The wind turbine power output, P_h^{WT}, is a function of the rated power and wind speed, which can be explained as follows [31]:

$$P_h^{WT} = \begin{cases} 0 & v_{co} < v_h < v_{ci} \\ {v_h}^3 \left(\frac{P_r}{v_r^3 - v_{ci}^3}\right) - P_r \left(\frac{v_{ci}^3}{v_r^3 - v_{ci}^3}\right) & v_{ci} \le v_h < v_r \\ P_r & v_r \le v_h < v_{co} \end{cases} \tag{12}$$

The wind turbine does not generate when the cut-in speed v_{ci}, necessary to start generating energy is not attained. Consequently, when wind speed surpasses the cut-off speed v_{co}, the generator halts its operation to protect its components.

2.5.2.3. Hydrogen chain: production, storage and consumption. Equation (13) express hydrogen production in a PEM electrolyzer [23]:

$$Q_h^{ELZ} = \frac{\eta_{ELZ} \times P_h^{ELZ}}{2F \times V_{rev}} \tag{13}$$

The efficiency of a PEM electrolyzer (η_{ELZ}) is the product of voltage efficiency η_V, Faradaic efficiency and auxiliary efficiency, representing the overall system efficiency in converting electrical energy into hydrogen [52]. A detailed calculation of the electrolyzer's efficiency is given in the supplementary file (S2).

A Proton Exchange Membrane (PEM) fuel cell generates electricity by combining hydrogen and oxygen, producing water as the only byproduct. The output power generated from the fuel cell is estimated as follows [53]:

$$P_h^{FC} = V_{cell} I_{cell_h} N_{fc} \tag{14}$$

A detailed calculation of the output voltage of a single cell (V_{cell}) is given in the supplementary file (S1). In the above equation, fuel cell current, I_{cell_h}, is a function of hourly hydrogen consumption in the fuel cell.

The amount of hydrogen consumed by the fuel cell is estimated as follows [53]:

$$Q_h^{FC} = \frac{P_h^{FC}}{2F.V_{cell}.N_{fc}} \tag{15}$$

Equation (15) expresses hydrogen storage dynamics, demonstrating how the level of hydrogen in the storage tank (H_h) changes over time [54].

$$H_h = H_{h-1} + Q_h^{ELZ} - Q_h^{FC} - G_h^H \qquad H_{min} < H_h < H_{max} \tag{16}$$

G_h^H denotes the surplus amount of hourly hydrogen available for selling to the market.

2.5.2.4. Biomass power generator. In this study, a steam Rankine cycle – biomass gasifier is used to integrate a biomass gasifier, boiler, steam turbine, condenser, and pump. Biomass is gasified to produce syngas, which is then combusted to release thermal energy for heating water in the boiler, generating steam that drives a turbine for electricity production [55]. Equations (17) and (18) represent the overall gasification reaction and combustion expressions respectively [56].

$$C_xH_yO_z + O_2 \rightarrow CO + H_2 + CO_2 + H_2O + Char \tag{17}$$

$$\dot{n}CO + \dot{m}H_2 + \frac{\dot{n} + \dot{m}}{2} O_2 \rightarrow \dot{n}CO_2 + \dot{m}H_2O \tag{18}$$

The power generation by the biomass plant P^{BM}, is formulated as a function of the amount of heat transferred to the boiler (Q_h^B) resulting from the gasification of dry biomass feed rate B_h, and thermal efficiency of the cycle (η_{st}), as follows:

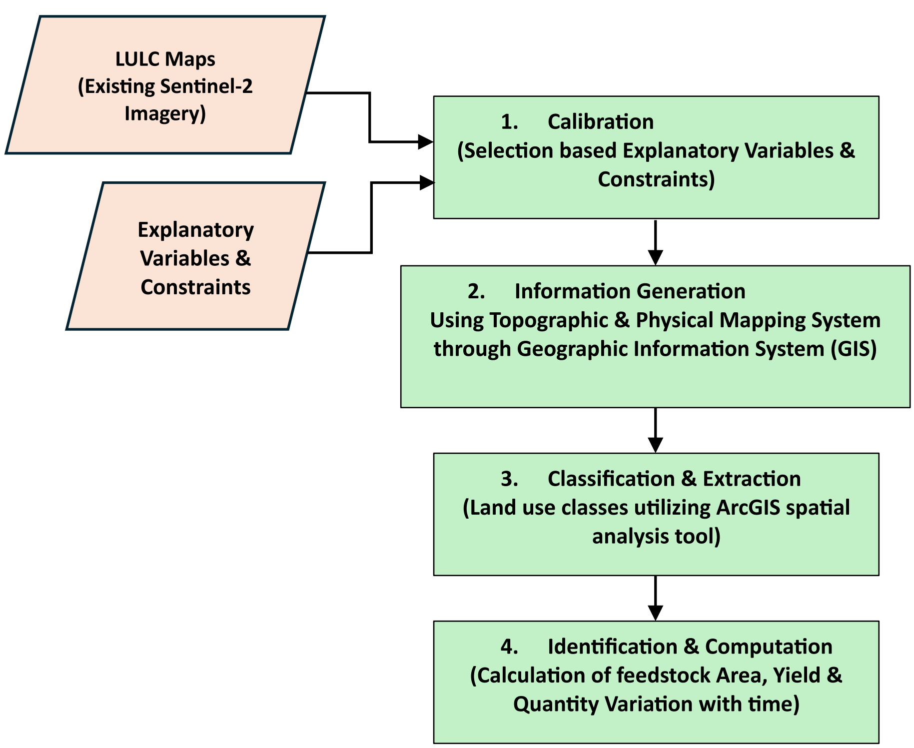


Fig. 5. LULC Mapping model for determination of feedstock fortitude.

$$P_h^{BM} = Q_h^B \cdot \eta_{st} \tag{19}$$

$$Q_h^B = B_h.LHV.(1 - e_{Ls}) \tag{20}$$

$$e_{Ls} = e_{fm} + e_{uc} + e_{dg} + e_{lh} + e_{ma} + e_{mfc} \tag{21}$$

The heat loss (e_{Ls}) is calculated from the losses attributed to fuel moisture, unburned fuel, dry flue gas, latent heat, moisture in the air, and various other manufacturing losses [57].

The value of HHV is estimated using the Gibbs free energy minimization model developed in this study. This approach minimizes the total Gibbs free energy of the biomass combustion reaction while ensuring elemental mass balance, using the Lagrange multiplier method, as described below [58]:

$$G_0 = \sum_{i=1}^{N} n_i \cdot G_i(T, P) \tag{22}$$

Here, $G_i(T, P)$ is the molar Gibbs free energy of species i, and n_i is its number of moles. The molar Gibbs free energy is expressed as:

$$G_i(T, P) = G_i^0(T) - RTln\left(\frac{n_i}{n_t}\right) \tag{23}$$

Substituting into the total Gibbs free energy expression:

$$G_0 = \sum_{i=1}^{N} n_i G_i^0(T) + \sum_{i=1}^{N} n_i RTln\left(\frac{n_i}{n_t}\right) \tag{24}$$

The elemental mass balance for each chemical element j, in the system is given by:

$$\sum_{i=1}^{N} a_{ij} n_i = b_j \tag{25}$$

To find the equilibrium composition, the Lagrangian function is formed and minimized using the Lagrange multipliers [59]:

$$\frac{\partial L}{\partial n_i} = \Delta G_{f,i}^o + n_i RTln\left(\frac{n_i}{n_t}\right) + \sum_{j=1}^{k} \lambda_j \left(\sum_{i=1}^{N_k} a_{ij} n_i - b_j\right) = 0 \tag{26}$$

Solving the above system yields the optimal mole numbers n_i for the product species i, the Lagrange multipliers λ_j, and satisfies the elemental balance constraints a_{ij}. Finally, the HHV value is estimated based on the appraised value of n_i, as follows [60]:

$$HHV = \sum_{i=1}^{N} HHV_i \frac{n_i}{n_t} \tag{27}$$

Table 2
H2-HRES component cost breakdown.

Component	Capital Cost ($/kW)		O&M Costs	
PV Panel	900	[69]	55 $/y	[21]
Wind Turbine	1200	[67]	41.78 $/y	[70]
Electrolyzer	890	[71]	20 $/y	[72]
Fuel Cell	1000	[73]	0.01 $/h	[21]
Hydrogen tank	1100	[74]	0	[72]
Biomass generator	600	[75]	15 $/y	[76]

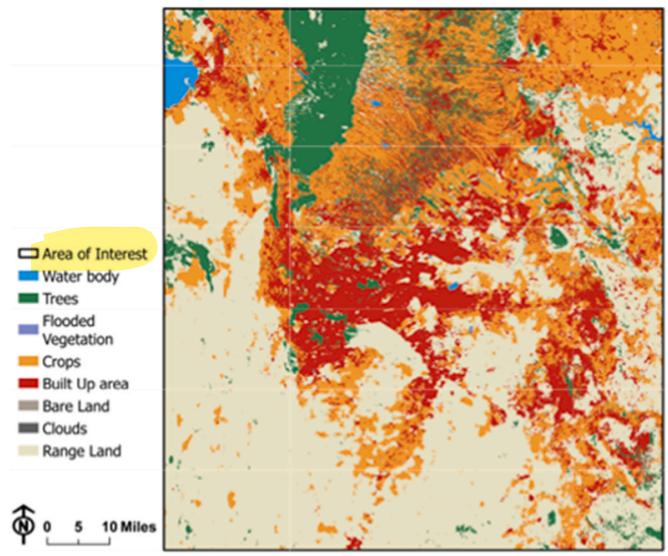


Fig. 7. LULC map classification of the area study.

The *LHV* is determined by taking away from the *HHV* energy lost as latent heat from water vapor produced during combustion [61].

$$LHV = HHV - l^{v} \cdot [W_r(1 + 8.94 \cdot C_{H_2})] \tag{28}$$

Table 3
Annual quantity of biomass feedstock.

Feedstock	Quantity (ton/year)
Dry Grass pallets	19,315,584
Wood pallets	1,622,350

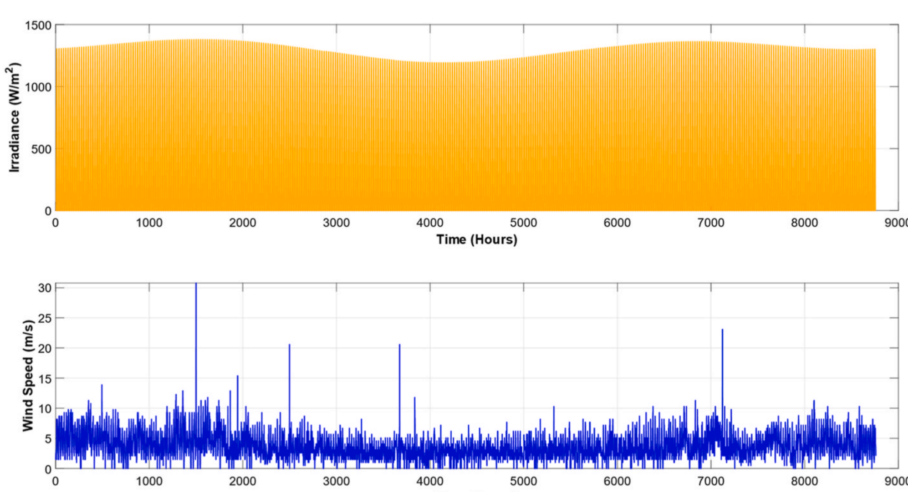


Fig. 6. Annual solar irradiance and wind speed in the selected study area.

Table 4
Land cover classification by percentage.

Land Use/Land Cover	Area (ha)	%
Flooded Vegetation	159	0.011
Bare Lands	394	0.026
Cloud	1005	0.067
Water Body	7135	0.472
Trees	102917	6.810
Built-Up area	288475	19.089
Crops	364909	24.147
Range Land	746188	49.378

Where, l^v represents the latent heat of vaporization of water, C_{H_2} refers to the hydrogen content of the feedstock and W_r is its moisture content [62].

The overall structure of the developed model with respect to the dispatch scheduling control logic introduced in section 2.5.2.1 is depicted in Fig. 4.

2.5.2.5. Biomass feedstock availability. In this study, the recoverable potential biomass feedstock amount is estimated using a Geographical Information System (GIS)-based Land Use Land Cover (LULC) mapping with a Bayesian classification approach [63]. This represents the portion of biomass that can be sustainably collected for energy use from the identified land categories. The overall process is described in Fig. 5.

The feedstock amount is quantified by classifying land cover types and estimating their respective areas from remotely sensed data [63]. Image pixel likelihoods $P_{(x_i|c)}$ in equation (29), estimate land cover classes c, and classified areas A, are computed using the Shoelace expression (30). Biomass feedstock fortitude B_{fd_r}, is calculated based on

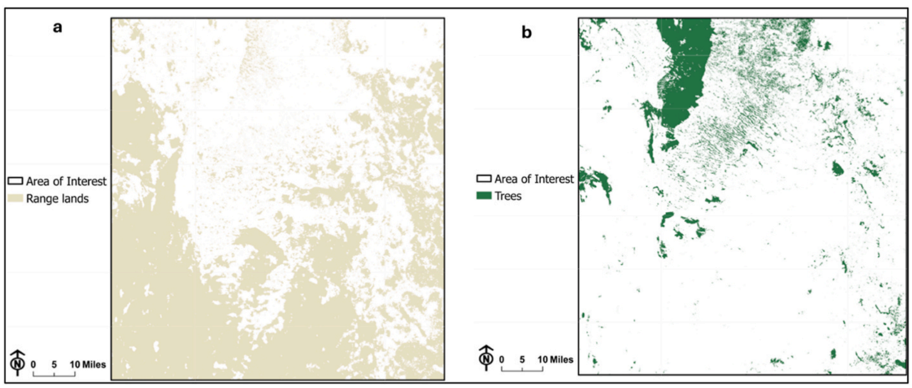


Fig. 8. Rangeland and Forested Land Classifications; a) distribution of feedstock in range land b) dispersal of trees in forested land.

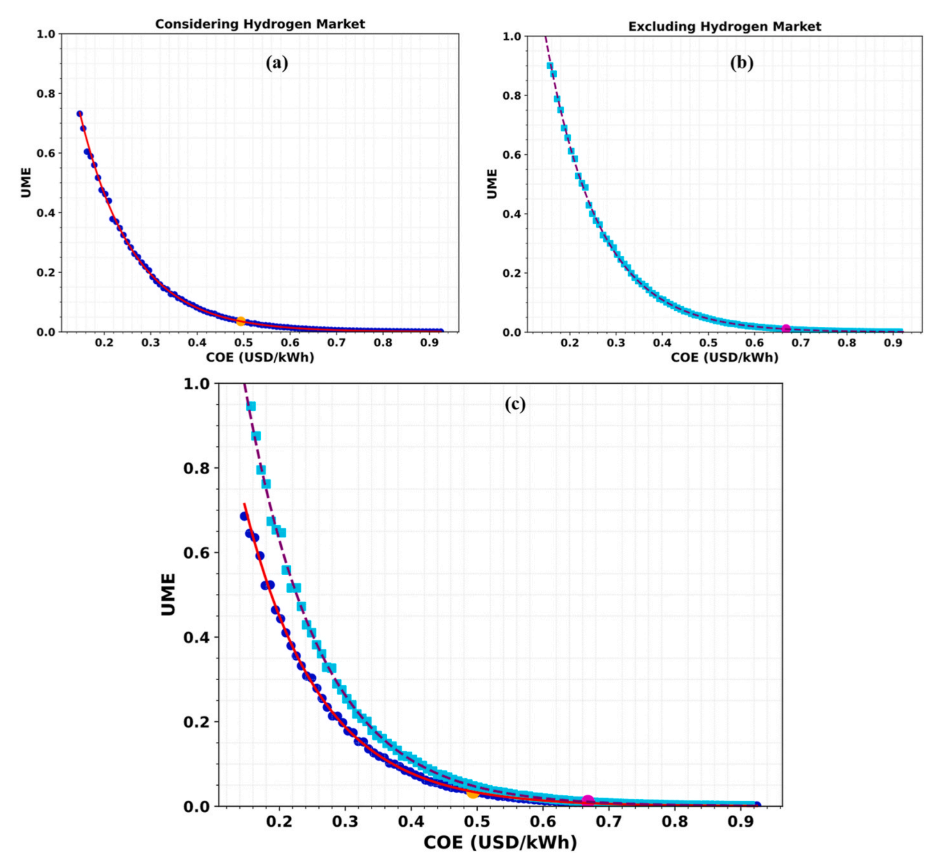


Fig. 9. Pareto curves representing the optimal configuration of the proposed H2-HRES, a) with the hydrogen market b) without the hydrogen market participation and c) combined Pareto curves.

wood pellet and grass yields Y, from classified forest and rangelands, using existing literature for yield estimation using equation (31) [64].

$$C_L = arg_{max_c}\left(P_c \prod_{i=1}^{n_c} P_{(x_i|c)}\right) \tag{29}$$

$$A_{F|R} = \frac{1}{2}\sum_{i=1}^{n_c}(x_i y_{i+1} - x_{i+1} y_i) \tag{30}$$

$$B_{fd_T} = B_{fd_F} + B_{fd_R} = \{(A_F \cdot Y_F) + (A_R \cdot Y_R)\} \tag{31}$$

The amount of feedstock available for the biomass generator is constrained by the estimated feedstock fortitude from the above method:

$$\sum_{h=1}^{8760} B_h \leq B_{fd_T} \tag{32}$$

3. Results and discussions

3.1. Area of study

Location

The proposed H2-HRES is applied for off-grid electrification of a developing suburb in Nairobi city, Kenya, located at 1° 19′ 14.33″ S, 36° 53′ 38.1″ E. The peak irradiance at the proposed site is 1.29 kW/m^2 with an hourly average of 0.479 kW/m^2 [65], while the peak and average wind speeds are 11.3 m/s and 3.69 m/s [66], respectively, as shown in Fig. 6. The total annual electrical load demand considered in this study is 127.8 MWh for three 7-story apartment buildings hosting a total of 70 households. The peak hourly electrical demand is witnessed in the morning period, 8–11 a.m., and evening hours, 8–10 p.m.

Table 2 presents the capital investment and annual operations and maintenance (O&M) costs of each component of the proposed H2-HRES. The capital and O&M costs of the wind turbine system are estimated based on the economic analysis of Kenya's recent Lake Turkana Wind Power (LTWP) project [67]. A local discount rate of 12 % is assumed [68]. The total capital cost is annualized using the Capital Recovery Factor (CRF) and subsequently converted into a unit cost (USD/kWh) by dividing by the system's annual energy output. Similarly, the annual O&M cost is normalized by the annual energy output to derive the corresponding unit cost in USD/kWh.

3.2. Biomass resource availability in the selected study area

The land cover classification within a 60 km radius centered on the area of study is analyzed using LULC mapping with spatial analysis, as shown in Fig. 7. This area reflects the realities of Nairobi's urban expansion, where biomass must be collected from outlying regions due to limited land availability within the city's immediate vicinity.

The approximate annual amount of dry grass and wood pallets available for biomass feedstock in land classified as rangeland and forested land is reported in Table 3. Table 4 rationalizes the details of the land-use divisions. Fig. 8 shows the two major divisions presenting the highest abundance of biomass feedstock: rangeland (a) and forested lands (b).

Table 5
Optimal configuration of the proposed H2-HRES.

Component	With the hydrogen market @ $6.6/kg
PV System	41.8 kW
Wind Turbines	30.1 kW
Biogenerator	27.4 kW
Fuel Cell	15.1 kW
Electrolyzer	40.3 kW

Table 6
Scenario definition.

Scenario	Characteristic			Climatic Condition
1	High LHV feedstock	High Irradiance	High Wind Speed	Dry season
2	High LHV feedstock	High Irradiance	Low Wind Speed	
3	High LHV feedstock	Low Irradiance	High Wind Speed	
4	High LHV feedstock	Low Irradiance	Low Wind Speed	Cold/Dry Season
5	Low LHV feedstock	High Irradiance	High Wind Speed	Rainy Season
6	Low LHV feedstock	High Irradiance	Low Wind Speed	
7	Low LHV feedstock	Low Irradiance	High Wind Speed	
8	Low LHV feedstock	Low Irradiance	Low Wind Speed	Cold/Rainy Season

3.3. Optimal configuration of the proposed H2-HRES, with and without the hydrogen market participation

Fig. 9 presents the Pareto front generated from the results of the multi-objective optimization model. The Pareto front illustrates the intricate trade-off between the competing objectives of COE and reliability. Achieving a favorable balance is challenging, as enhancing one objective often results in a degradation of the other; specifically, reducing the COE undermines the reliability index, and vice versa. The optimization process iterates on an hourly basis throughout the entire year, systematically identifying the optimal values for $f_1(COE)$ and $f_2(UME)$. This iterative approach ensures a robust compromise, minimizing energy costs while maintaining system reliability, thereby preventing power outages in the proposed H_2-HRES.

In Fig. 9 (a), the system achieves an optimal reliability index of 0.961 at a COE of $0.494/kWh, including hydrogen trading in the market at about $6.6/kg [77]. Conversely, in Fig. 9 (b), where hydrogen market trading is excluded, the optimal reliability index increases to

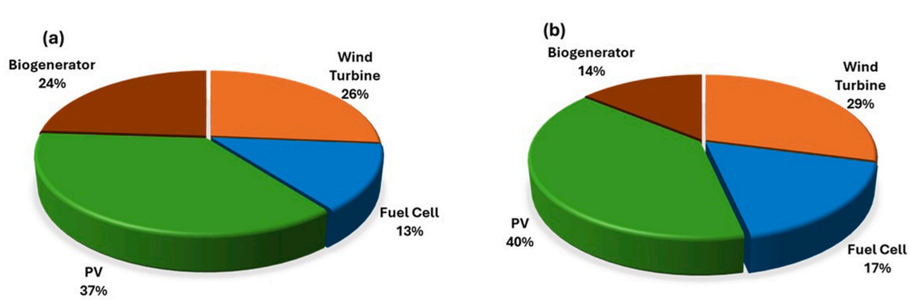


Fig. 10. Optimal installed capacity of the proposed H2-HRES, a) with the hydrogen market and b) without the hydrogen market participation.

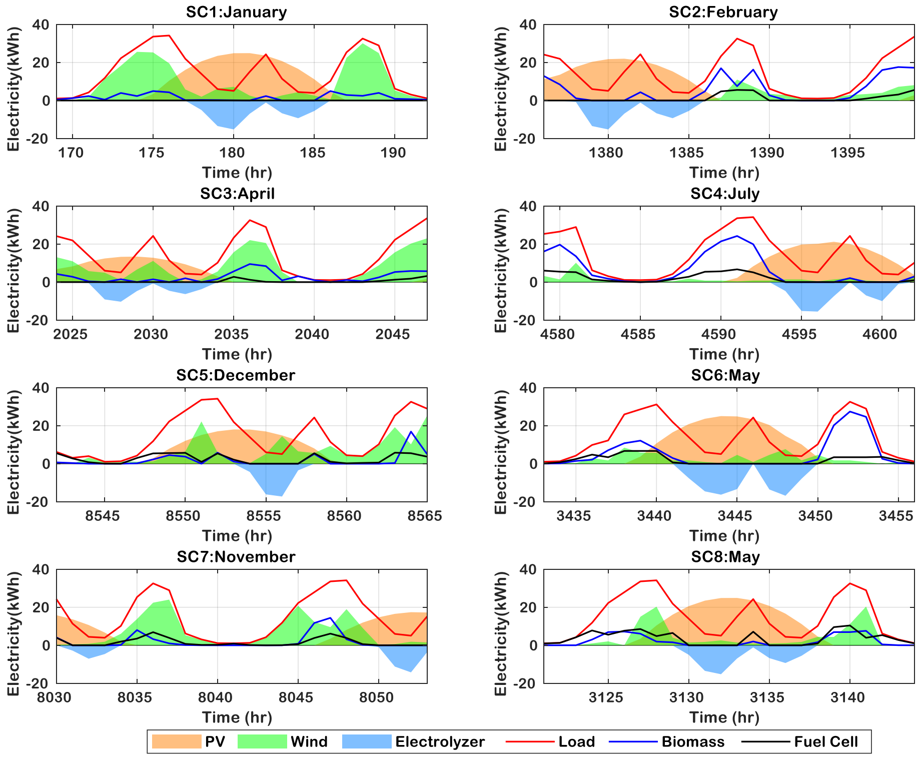


Fig. 11. Electricity demand-supply in the different scenarios with hydrogen market participation.

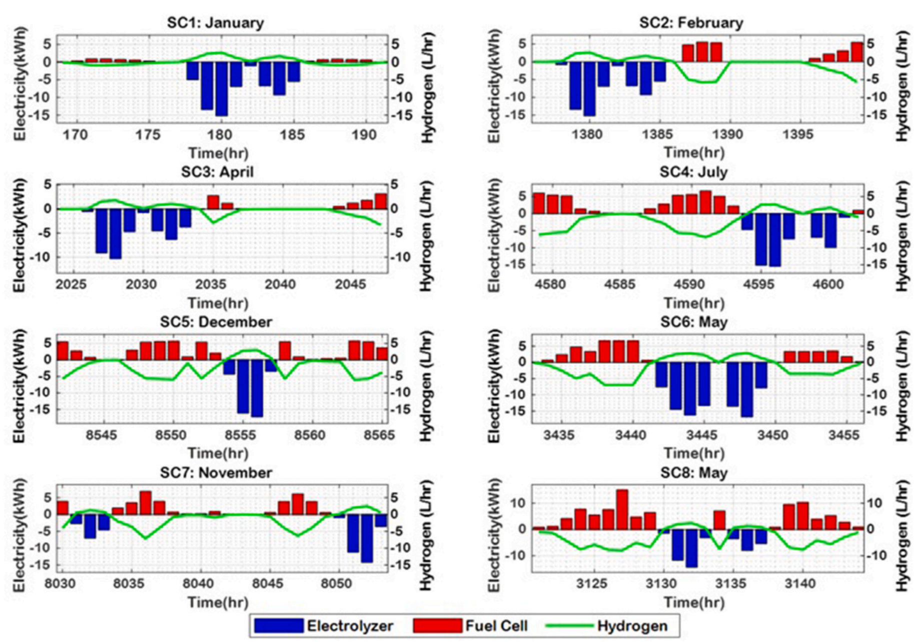


Fig. 12. Hydrogen production and consumption in different scenarios with hydrogen market participation.

0.978, but at a higher COE of \$0.668/kWh. The insights underscore the economic advantage of participating in the hydrogen market, demonstrating that selling surplus hydrogen reduces the COE by about \$0.174/kWh, effectively lowering consumer energy expenditures while maintaining system reliability.

Furthermore, the analysis reveals that, increased utilization of the fuel cell enhances system reliability by ensuring a more stable power supply. However, this reliability improvement comes at the expense of a higher COE due to increased hydrogen consumption. Participation in the hydrogen market significantly influences the optimal system configuration. This will lead to a boost in hydrogen trading in the market and reduce COE. In the absence of hydrogen market participation, the system relies more heavily on hydrogen for direct electricity generation. This shift emphasizes the economic trade-offs between hydrogen market participation and hydrogen power generation, when optimizing the HRES configuration shown in Fig. 10.

The optimal configuration of the proposed HRES is given in Table 5.

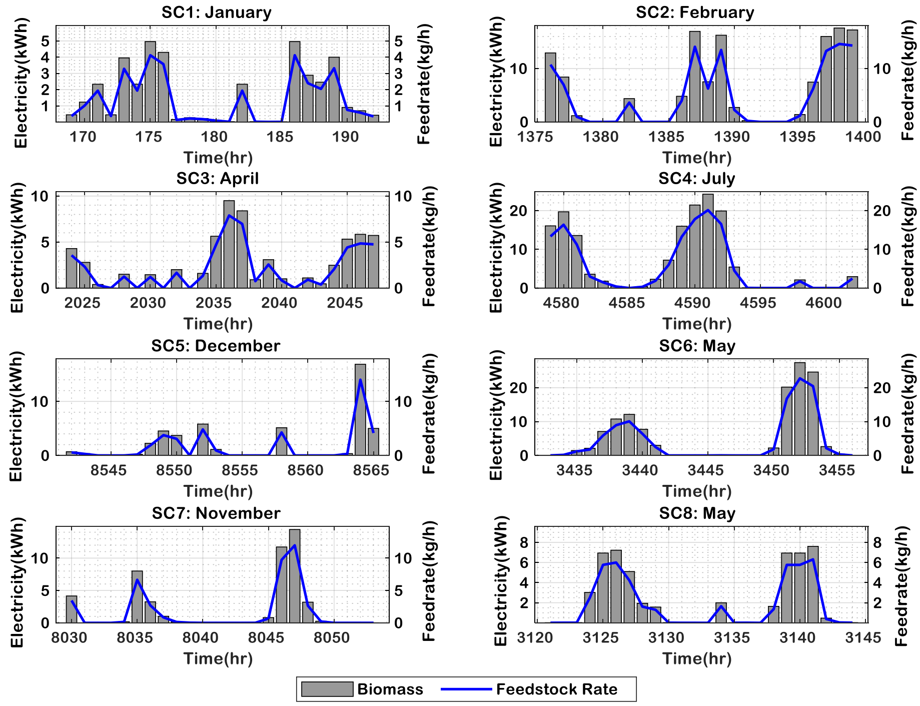


Fig. 13. Biomass power generation and feed-rate variability with hydrogen market participation.

3.4. Scenario analysis

The approach to scenario analysis rigorously incorporates diverse factors of environmental interactions and potential weather variations reflected in irradiance levels, wind speeds, and biomass feedstock LHV characteristics. Table 6 shows the eight identified scenarios that lie within the two dominant climatic conditions experienced in Kenya: rainy and dry seasons.

A blend of wood pellets and grass mixed in the ratio of their prevalence, 1:10, is used in the determination of the final values of LHV throughout the different seasons of the year. The LHV of the blended feedstock reaches 14.3 MJ/kg, when precipitation reaches 120 mm. In contrast, the highest LHV of 17.5 MJ/kg occurs when precipitation is at its lowest (0–12 mm), typically recorded in February and September.

Fig. 11 represents the results of the scenario analysis. In Scenario 1 (SC1:January), wind and solar power generation dominate due to favorable wind speeds and high solar irradiance levels. The system prioritizes the biomass generator over the fuel cell as a complementary power source, leveraging its feedstock's higher LHV and choosing to sell the produced hydrogen instead of using it for electricity generation.

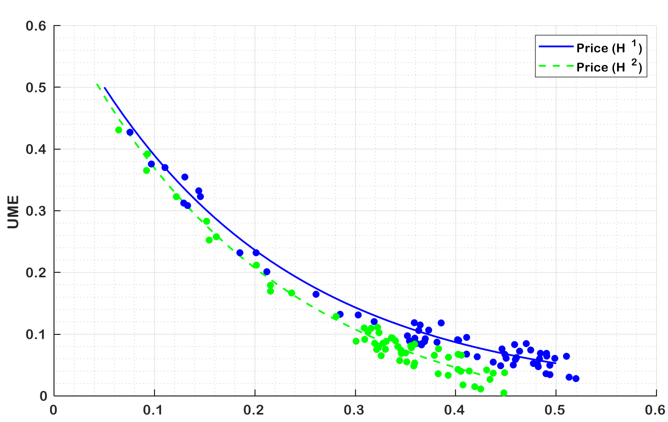


Fig. 14. Impact of hydrogen market price on the COE and UME.

Table 7
Optimal configurations under different hydrogen market prices.

Component	H^1 = \$6.6/kg	H^2 = \$9.9/kg
PV System	41.8 kW	43.2 kW
Wind Turbine	30.1 kW	29.4 kW
Biogenerator	27.4 kW	30.3 kW
Fuel Cell	15.1 kW	13.5 kW
Electrolyzer	40.3 kW	32.7 kW

Scenarios 2, 3, and 4 exhibit a greater dependence on biomass generation, with the system favoring hydrogen sales over fuel-cell-based electricity production. This unit commitment strategy used in the model is primarily influenced by hydrogen revenue impacts on the COE and the availability of high-quality biomass feedstock, indicated by elevated LHV values. In Scenario 4 (July), lower wind speeds necessitate an increased reliance on the biomass generator, particularly for meeting peak nighttime electricity demand, hence recording a peak output of 24.3 kWh. In contrast, Scenarios 5, 6, 7, and 8 are characterized by rainy weather conditions, leading to lever biomass feedstock LHV values. As a result, the system is moving towards increased fuel cell usage for electricity generation due to the limited potential of biomass energy. During these periods, less hydrogen is available for market sale, as a higher fraction is allocated for power generation. Notably, Scenario 8 (May) records the highest fuel cell utilization throughout the entire optimization period. Across all scenarios, the proposed H2-HRES strategically directs excess electricity generated during solar hours to the electrolyzer, ensuring optimal hydrogen production for both market participation and power generation.

Fig. 12 illustrates the comparative analysis of hydrogen generation

Table 8
Weather-based hydrogen market price sensitivity analysis.

	Dry season		Wet/rainy season	
Hydrogen price	H^1 = \$6.6/kg	H^2 = \$9.9/kg	H^1 = \$6.6/kg	H^2 = \$9.9/kg
COE	\$0.452/kWh	\$0.398/kWh	\$0.511/kWh	\$0.442/kWh

Table 9
Validation of the current study with existing similar works.

Study	Location	System Configuration	LCOE ($/kWh)	Key Inference
[15]	Riyadh, Jeddah, Dhahran, Saudi Arabia	PV/Wind/FC/H_2	1.21	Hydrogen-based PV-Wind systems provide a fully renewable, battery-free solution but remain less cost-effective due to high capital costs, underscoring the need for hydrogen technology cost reductions.
[17]	South Australia	PV/Wind/Battery/H_2	0.63	Hybrid hydrogen-battery storage significantly reduces electricity costs and enhances reliability, proving optimal for large-scale, long-term renewable energy integration.
[21]	South Africa	PV/Wind/Fuel Cell/ H_2	2.34	The microgrid system is viable over grid extension beyond 8.81 km for rural clinic electrification.
[78]	Global-scale, standalone (off-grid) system	Wind/Electrolyzer/ FC/Battery	0.69	The ABC-optimized wind-hydrogen system offers a cost-effective and reliable off-grid energy solution, achieving a COE of $0.6908/kWh and improving performance through strategic cost reductions and turbine selection.
[79]	Sagar, Central India	PV/Wind/FC/ Battery/Diesel	0.51	Utilizes game theory optimization, yielding 99 % RE penetration with a cost-effective hybrid system, demonstrating strategic use of H_2 for sustainable and resilient power generation
[80]	China (Remote area)	Wind/Electrolyzer/ FC/H_2	0.59–0.94	Optimized wind-hydrogen system offers 100 % reliability, but the cost varies with allowable load loss.
[This study]	Nairobi, Kenya	PV/Wind/Biomass/ FC/H_2	0.49	H_2 trading reduces COE; seasonal and reliability trade-offs are identified

via electrolysis versus fuel cell consumption across eight scenarios, highlighting the dynamic trade-off between production and utilization. In Scenarios 1, 2, and 3, high renewable energy availability driven by strong wind speeds and elevated solar irradiance facilitates greater electrolyzer output. Consequently, hydrogen trading is prioritized over electricity generation by the fuel cell. On the contrary, in Scenarios 5, 7, and 8, wet conditions lead to reduced biomass LHV, shifting system reliance toward fuel cell utilization. Scenario 8 exhibits the highest peak fuel cell usage, significantly limiting hydrogen availability for market sale. Therefore, hydrogen generation and consumption in HRES fluctuate with seasonal resources and optimization priorities. High irradiance boosts electrolyzer output, enhancing hydrogen sales, while low wind speed and biomass LHV in wet conditions increase fuel cell reliance.

Biomass power generation and feedstock consumption across the eight scenarios exhibit strong seasonal variability, directly impacting COE and system reliability. In Fig. 13, Scenarios 1–4 align with high LHV feedstock availability, enhancing biomass power generation and reducing fuel cell reliance. This increases hydrogen sales in the market, lowering COE while sustaining reliability. Scenario 4, with a peak biomass output of 25 kWh and lower wind availability, shifting peak-load supply to biomass and raising feedstock consumption to 20 kg/h. Conversely, Scenarios 5–8 highlight the impact of wet conditions, reducing biomass LHV and necessitating higher feedstock input to maintain power generation. This inefficiency elevates COE due to lower biomass output and increased dependence on the fuel cell. Ultimately, biomass variability dictates H_2-HRES cost-effectiveness, requiring optimal feedstock utilization to balance hydrogen trade, fuel cell reliance, and system reliability.

3.5. *Impact of hydrogen market price*

The sensitivity analysis is conducted to evaluate the impact of hydrogen market price on the optimal COE of the proposed H_2-HRES, as shown in Fig. 14. Two market prices are considered here: $H^1 = \$6.6/kg$ and $H^2 = \$9.9/kg$ [45]. Under H^1, the optimal COE is determined to be $0.494/kWh, whereas under H^2, it declines to $0.405/kWh. This reduction of $0.089/kWh highlights the inverse correlation between hydrogen pricing and electricity costs, emphasizing the potential economic advantage of higher hydrogen market participation in enhancing system's economic viability.

As hydrogen price increases, the model prioritizes market participation over direct electricity generation from hydrogen,. The fuel cell capacity decreases (15.1 kW– 13.5 kW) since hydrogen is redirected to the market rather than used for on-site power generation. To compensate for reduced fuel cell use, biomass capacity increases (27.4 kW–30.3 kW), ensuring reliability. Additionally, PV panel installations nearly doubled to enhance renewable energy capture, supporting electrolyzer operation. The detailed optimal configuration of the proposed H_2-HRES under different hydrogen market prices is given in Table 7.

To further detail analysis of the impact of hydrogen market price, A weather-based sensitivity analysis is conducted to assess the influence of seasonal variations on the optimal COE under two aforementioned hydrogen market prices, during both the dry and wet/rainy seasons. As depicted in Table 8, the optimal COE exhibits a marked shift, decreasing from a peak of $0.511/kWh during the rainy season to a minimum of $0.452/kWh in the dry season. Similarly, seasonal fluctuations affect the optimal COE, which drops from $0.442/kWh during the rainy season to $0.398/kWh in the dry season. These findings underscore the significant role of climatic conditions in optimizing the cost structure of electricity generation, demonstrating how weather variability influences both the unit commitment and cost reduction strategies.

3.6. *Model validation*

Table 9 presents a comparative validation of the current hydrogen-based hybrid renewable energy system (H_2-HRES) against existing similar studies conducted globally. The results highlight that hydrogen integration into hybrid systems is increasingly seen as a viable path toward sustainable, off-grid electrification. However, the economic performance varies widely depending on configuration, technology maturity, and contextual factors such as location and resource availability. The present study, conducted in Nairobi, Kenya, achieves the lowest COE among the referenced systems at $0.494/kWh. This result is attributed to strategic participation in the hydrogen market, which allows for effective cost trading, as well as the integration of biomass with PV, wind, and fuel cell technologies. In comparison, other studies, such as the Saudi Arabia-based output (COE = $1.208/kWh), demonstrate higher costs due to the exclusion of market dynamics and reliance solely on capital-intensive components like electrolyzers and fuel cells. Studies in South Australia and China emphasize hybrid storage with batteries, which helps reduce COE to as low as $0.626/kWh, showing the cost advantage of hybridized synergy. Meanwhile, global and Indian systems that incorporate game theory and multi-source configurations report moderate COEs between $0.505 and $0.6908/kWh. However, systems in rural South Africa remain costlier ($2.34/kWh), constrained by grid extension thresholds. In a nutshell, the validation confirms that integrating hydrogen market mechanisms into HRES designs substantially improves cost optimization. This reinforces the notion that policy, market readiness, and strategic system configurations are as vital as technology itself in achieving viable, low-cost, hydrogen-powered energy systems.

4. Study limitations

Despite the comprehensive approach adopted in this study, several limitations remain. First, the model assumes the existence of a functional local hydrogen market in Nairobi; however, actual hydrogen demand, infrastructure, and market maturity in Kenya are still in nascent stages. The hydrogen sales dynamics are based on projections rather than current empirical market data. Secondly, land-use competition between PV deployment and biomass feedstock availability is not clearly constrained in the model to reflect real-world spatial limitations during optimization. Additionally, the model does not explicitly account for replacement cycles of critical components such as electrolyzers and fuel cells, which could impact system economics and reliability throughout the project's lifetime. Finally, the study does not incorporate evolving policy frameworks, subsidies, or carbon credit mechanisms, all of which could significantly influence hydrogen economics and renewable energy deployment in Kenya.

5. Conclusion

This study presents a comprehensive techno-economic optimization of a hydrogen-based HRES, using a multi-objective approach, systematically balancing cost minimization and reliability enhancement. The Pareto front analysis highlights the trade-offs between COE and system reliability, revealing that the hydrogen market participation significantly reduces COE, while only marginally affecting reliability. The optimization results indicate that monetizing surplus hydrogen leads to a COE reduction of \$0.174/kWh, while maintaining a reliability index of 0.961. However, in the absence of hydrogen market trading, the reliability index enhances to 0.978, though at a higher COE of \$0.668/kWh. The HRES configuration is strongly influenced by hydrogen market participation, where hydrogen sales in the market increase PV and wind contributions, respectively, while reducing the fuel cell's role and biomass. Seasonal variability further shapes system operations. In dry seasons, higher solar and wind availability favors hydrogen production via electrolysis, promoting hydrogen sales over internal consumption. During wet seasons, biomass's lower LHV declines, necessitating increased fuel cell utilization, which shifts the system toward greater hydrogen consumption for direct electricity generation. Wind variability also impacts biomass dependence, with lower wind speeds expanding the need for biomass power to sustain peak-load demand. Hydrogen generation and fuel cell utilization follow seasonal trends, with electrolyzer output peaking during high renewable energy availability, while wet conditions require higher fuel cell consumption, limiting hydrogen availability for market participation. These fluctuations directly influence hydrogen market participation, where higher hydrogen production in dry seasons correlates with lower COE, while wet conditions elevate COE due to increased hydrogen power generation. The results revealed that biomass feedstock plays a crucial role in the economic viability of the proposed H2-HRES, as high LHV in dry seasons supports increased power generation, reduces reliance on fuel cells, and enhances hydrogen sales in the market. Conversely, wet seasons reduce biomass energy efficiency, increasing feedstock demand, raising COE, and necessitating greater fuel cell reliance. Hydrogen market price sensitivity further affects system economics, where higher hydrogen prices (\$9.9/kg) lower COE from \$0.494/kWh to \$0.405/kWh by favoring hydrogen commercialization over direct fuel cell usage. The system adapts by increasing electrolyzer capacity, reducing fuel cell size, and expanding PV installations, ensuring economic viability while slightly reducing reliability. Ultimately, this study underscores the economic and operational trade-offs between hydrogen market participation and power generation, demonstrating that hydrogen market participation enhances economic feasibility by reducing COE, albeit with minor reliability trade-offs. Seasonal variations in renewable resource availability and biomass quality further emphasize the need for adaptive operational strategies. These insights are particularly valuable for policymakers, energy planners, and investors aiming to implement sustainable and cost-effective energy solutions in both urban and off-grid contexts. While the model is applied to the specific case of Nairobi, its modular architecture, multi-objective optimization framework, and flexible input parameters make it broadly applicable to similar electrification challenges across Sub-Saharan Africa and other comparable regions.

The results demonstrate that hydrogen market participation can substantially reduce COE with minimal reliability loss, reinforcing the need for Kenya to establish a transparent, regulated local hydrogen market, initially targeting urban industrial and agricultural consumers. Seasonal variability in renewable resources calls for policies that enable flexible, market-linked dispatch strategies in hybrid renewable–hydrogen systems. The proven performance advantage of PV/Wind/Biomass/Hydrogen configurations over single-resource systems supports targeted incentives for hybrid energy projects. Urban electrification strategies should also incorporate hydrogen economics, recognizing its dual role as an energy storage medium and a revenue source through market participation.

Building on the findings of this study, future research will focus on incorporating stochastic or agent-based modeling of hydrogen market development, capturing demand fluctuations, price elasticity, competition with alternative energy carriers, and supply chain logistics to enhance model realism. Further work will also broaden the scope of uncertainty analysis, considering variations in both technical and economic parameters to provide a more comprehensive view of system robustness. Additionally, future work will integrate electrolyzer and fuel cell degradation profiles, along with replacement cycles, to enable a more comprehensive long-term economic and operational assessment. Finally, the model will be applied to multiple geographic regions and scaled for interconnected microgrids to evaluate the broader applicability and economic potential of hydrogen-based HRES systems across Sub-Saharan Africa.

Nomenclature

HRES	Hybrid renewable energy systems	UME	Unmet Energy
COE	Cost of electricity (\$/kWh)	HHV	Higher heating value (kWh/kg)
PEM	Proton exchange membrane	$\Delta G^{o}_{f,i}$	Standard Gibbs free energy (J/mol) of i
LULC	Land use land cover	n_t	Total number of moles for species
GIS	Geographic information system	R	Universal gas constant (J/mol·K)
H_2-HRES	Hydrogen-based HRES	P	Gas Pressure (Pa)
P_h^{FC}	Fuel cell generation (kWh)	$G_i(T, P)$	Gibbs free energy (J/mol) of species i
P^{HRES}	Total HRES generation (kWh)	$G_i^0(T)$	Standard Gibbs free energy (J/mol) of species i
TC	Total Cost (\$)	L	Lagrange function
MILP	Mixed-Integer Linear Programming	λ_j	Lagrange multiplier
V_{cell}	Single cell voltage (V)	k	Number of conserved elements
F	Faraday's constant (C/mol)	N_k	Number of chemical species
N_{fc}	Number of cells (A)	HHV_i	HHV (kWh/kg) of component i
I_{cell}	Stack current	C_x	Carbon atoms
P_h^{ELZ}	Electrolyzer consumption (kWh)	H_y	Hydrogen atoms
V_{rev}	Reversible voltage (V)	O_z	Oxygen atoms
Q_h^{ELZ}	Hydrogen Output (kg/h)	O_2	Oxygen
η_{ELZ}	Electrolyzer efficiency	CO	Carbon monoxide
P_h^D	Electrical load demand (kWh)	CO_2	Carbon dioxide
Q_h^{FC}	Hydrogen consumed by fuel cell (kg/h)	H_2	Hydrogen
LHV	Lower heating value (kWh/kg)	$Char$	Solid carbon residue (kg)

(continued on next page)

(*continued*)

C_{H_2}	Mass fraction of Hydrogen	H_2O	Water
W_r	Moisture content (%)	C_L	Predicted land cover class
l^v	Latent heat of vaporization (kJ/kg)	$P_{(x_i\|c)}$	Pixel likelihoods
i	Chemical species index	$A_{F\|R}$	Classified feedstock area (ha)
j	Chemical elements index	(x_i, y_i)	Area coordinates (m)
a_{ij}	Number of atoms of j in species i	B_{fd_T}	Feedstock fortitude
b_j	Number of moles of element j	$Y_{F\|R}$	Forested/Range land yield
n_i	Number of moles species i	n_c	Input variables
p^H	Unit price of Hydrogen ($/kg)	P_c	Prior probability of class c
dr	Discount rate (yr^{-1})	B_{fd_F}	Feedstock fortitude in forested
n	Number of years	B_{fd_R}	Feedstock fortitude in range land
τ	Type of generator	A_F	Forested Area (ha)
λ	Unit cost of capital expense ($/kWh)	A_R	Range land Area (ha)
G^H	Hydrogen available for sale (kg)	Y_F	Forested yield (t/ha/yr)
H_h	Tank hydrogen level	Y_R	Range land yield (t/ha/yr)
P_h^{PV}	PV generation (kWh)	f_1	COE objective function ($/kWh)
PV_{max}	Maximum PV power rating (kW)	f_2	Reliability objective function
Pv_{df}	PV derating factor (0.8)	ϵ	Epsilon constraint
N_P	Number of PV modules	Ri	Reliability index
G_h	In-plane irradiance (W/m^2)		
T_{NOCT}	Normal operating cell temperature (°C)	P_h^{BM}	Biomass genration (kWh)
T_{ah}	Ambient temperature (°C)	η_{st}	Steam Rankine cycle efficiency
G_{STC}	Standard test condition Irradiance	Q_h^B	Heat transferred in the boiler
K_T	**Temperature coefficient** ($°C^{-1}$)	B_h	Biomass feed rate (kg/h)
P_h^{WT}	Wind turbine power output (kW)	e_{Ls}	Overall efficiency losses
P_r	Rated wind turbine power (kW)	e_{fm}	Fuel moisture losses
v_{co}	Cut-out wind speed (m/s)	T_c	Cell temperature (°C)
v_h	Wind speed (m/s)	e_{uc}	Unburned fuel losses
v_{ci}	Cut-in wind speed (m/s)	e_{dg}	Dry gas losses
v_r	Rated wind speed (m/s)	e_{lh}	Latent heat losses
H_{min}	Minimum H_2 level	e_{ma}	Moisture in air losses
β	Binary activation state	e_{mfc}	Manufacturing losses
$\beth_0, \beth_a, \beth_b, \beth_c$	Binary switching flags	H_{max}	Maximum H_2 level
N	Number of different gas species	G_0	Total Gibbs free energy (J)
λ_{PV}	PV capital costs ($/kWh)	$\lambda_{OM\tau}$	All system O&M costs ($/kWh)
λ_{WT}	Wind turbine costs (($/kWh)	$\lambda_{H_{st}}$	Hydrogen storage tank costs ($/kWh)
λ_{ELZ}	Electrolyzer costs ($/kWh)	λ_{FC}	Fuel cell costs ($/kWh)
$\dot{n}$	Number of moles of CO	$\dot{m}$	Number of moles of H_2 gas
S_0	instantaneous load situation	S_1	Surplus generation situation
S_2	insufficient generation situation	T	Gas temperature (K)

CRediT authorship contribution statement

Alphonce Ngila Mulumba: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Hooman Farzaneh:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2025.151474.

References

[1] Omer AM. Clean and green energy technologies, sustainable development, and environment. 2014. p. 287–320. https://doi.org/10.4018/978-1-4666-6252-0.ch015.

[2] Fields N, Ryves D, Yeganyan R, Cannone C, Tan N, Howells M. Evidence-based policymaking: insights and recommendations for the implementation of clean energy transition pathways for Kenya's power sector. Energies Dec. 2023;16(23):7904. https://doi.org/10.3390/en16237904.

[3] USAID. Kenya climate change country profile. Washington DC 20004, https://www.usaid.gov/climate/country-profiles/kenya#:~:text=Although Kenyacontributeslessthan,usualscenarioandinline. [Accessed 4 July 2024].

[4] Okendo EO, Wekesa CW, Saulo MJ. Optimal placement of Phasor Measurement Unit considering system observability Redundancy Index: case study of the Kenya power transmission network. Heliyon Jul. 2021;7(7):e07670. https://doi.org/10.1016/j.heliyon.2021.e07670.

[5] Ngila Mulumba Alphonce, Hooman Farzaneh. Economic contributions and green job implications of Japan's energy transition: an integrated assessment of decarbonization scenarios in Kyushu Region. Proc Int Exch Innov Conf Eng Sci Oct. 2024;10:1246–54. https://doi.org/10.5109/7323416.

[6] Barasa M, Bogdanov D, Oyewo AS, Breyer C. A cost optimal resolution for Sub-Saharan Africa powered by 100% renewables in 2030. Renew Sustain Energy Rev Sep. 2018;92:440–57. https://doi.org/10.1016/j.rser.2018.04.110.

[7] Zoulias PT, I E, Glockner R, Lymberopoulos N, Tsoutsos T, Vosseler I, Gavalda O, Mydske HJ. Integration of hydrogen energy technologies in stand-alone power systems analysis of the current potential for applications [Online]. Available: https://www.osti.gov/etdeweb/biblio/20457759; 2004.

[8] Li Z, Wu W, Si Y, Chen X. Optimal siting and sizing of hydrogen production modules in distribution networks with Photovoltaic uncertainties. Energies Nov. 2023;16(22):7636. https://doi.org/10.3390/en16227636.

[9] Khalil M, Dincer I. Investigation of a community-based clean energy system holistically with renewable and hydrogen energy options for better sustainable development. J Clean Prod Feb. 2024;440:140895. https://doi.org/10.1016/j.jclepro.2024.140895.

[10] El-Sattar HA, Kamel S, Sultan HM, Zawbaa HM, Jurado F. Optimal design of Photovoltaic, Biomass, Fuel Cell, Hydrogen Tank units and Electrolyzer hybrid system for a remote area in Egypt. Energy Rep Nov. 2022;8:9506–27. https://doi.org/10.1016/j.egyr.2022.07.060.

[11] Elegbeleye I, Oguntona O, Elegbeleye F. Green hydrogen: pathway to net zero Green House gas emission and global climate change mitigation. Hydro Apr. 2025;6(2):29. https://doi.org/10.3390/hydrogen6020029.

[12] Singh BJ, Sehgal R. "Green hydrogen production: bridging the gap to a sustainable. Energy Future 2024:83–124. https://doi.org/10.1007/978-981-97-1339-4_5.

[13] Ajagun AS, Mao W, Sun X, Guo J, Adebisi B, Aibinu AM. The status and potential of regional integrated energy systems in Sub-Saharan Africa: an Investigation of the feasibility and implications for sustainable energy development. Energy Strategy Rev May 2024;53:101402. https://doi.org/10.1016/j.esr.2024.101402.

[14] Ruth EB Mark F, Jadun Paige, Gilroy Nicholas. Connelly, "The Technical and Economic Potential of the H2@Scale Concept within the United States,". Newyork, https://www.nrel.gov/docs/fy21osti/77610.pdf; 2020.

[15] Al-Sharafi A, Sahin AZ, Ayar T, Yilbas BS. Techno-economic analysis and optimization of solar and wind energy systems for power generation and hydrogen production in Saudi Arabia. Renew Sustain Energy Rev Mar. 2017;69:33–49. https://doi.org/10.1016/j.rser.2016.11.157.

[16] Urs RR, Chadly A, Al Sumaiti A, Mayyas A. Techno-economic analysis of green hydrogen as an energy-storage medium for commercial buildings. Clean Energy Feb. 2023;7(1):84–98. https://doi.org/10.1093/ce/zkac083.

[17] Kharel S, Shabani B. Hydrogen as a long-term large-scale energy storage solution to support renewables. Energies Oct. 2018;11(10):2825. https://doi.org/10.3390/en11102825.

[18] Adedoja OS, Sadiku ER, Hamam Y. A techno-economic assessment of the viability of a photovoltaic-wind-battery storage-hydrogen energy system for electrifying primary healthcare centre in Sub-Saharan Africa. Energy Convers Manag X Jul. 2024;23:100643. https://doi.org/10.1016/j.ecmx.2024.100643.

[19] Ruben Zieba Falama SYD, Hamda Soulouknga Marcel, Dumbrava Virgil, Saidi Abdelaziz Salah, Kaoutoing Maxime Dawoua, Ben Salah Chokri. Renewable energy systems for hydrogen production in Sub-Saharan Africa: a comparative Study based on A techno-economic analysis. UPB Sci Bull 2023;85(1):2286–3540 [Online]. Available: https://www.researchgate.net/publication/369224754.

[20] Falama RZ, Saidi AS, Soulouknga MH, Ben Salah C. A techno-economic comparative study of renewable energy systems based different storage devices. Energy Mar. 2023;266:126411. https://doi.org/10.1016/j.energy.2022.126411.

[21] Ayodele TR, Mosetlhe TC, Yusuff AA, Ogunjuyigbe ASO. Off-grid hybrid renewable energy system with hydrogen storage for South African rural community health clinic. Int J Hydrogen Energy Jun. 2021;46(38):19871–85. https://doi.org/10.1016/j.ijhydene.2021.03.140.

[22] Zoulias EI, Lymberopoulos N. Techno-economic analysis of the integration of hydrogen energy technologies in renewable energy-based stand-alone power

systems. Renew Energy Apr. 2007;32(4):680–96. https://doi.org/10.1016/j.renene.2006.02.005.
[23] Khan T, Yu M, Waseem M. Review on recent optimization strategies for hybrid renewable energy system with hydrogen technologies: state of the art, trends and future directions. Int J Hydrogen Energy Jul. 2022;47(60):25155–201. https://doi.org/10.1016/j.ijhydene.2022.05.263.
[24] Zeng Y, Cai Y, Huang G, Dai J. A review on optimization modeling of energy systems planning and GHG emission mitigation under uncertainty. Energies Oct. 2011;4(10):1624–56. https://doi.org/10.3390/en4101624.
[25] Serrano-Arévalo TI, Tovar-Facio J, Ponce-Ortega JM. Optimal incorporation of intermittent renewable energy storage units and green hydrogen production in the electrical sector. Energies Mar. 2023;16(6):2609. https://doi.org/10.3390/en16062609.
[26] Bouramdane A-A. Optimizing energy dispatch strategies for hydrogen-based hybrid systems to address the camel-duck shape of electricity demand with increasing solar photovoltaic penetration. Int J Hydrogen Energy Mar. 2024;110:1–31. https://doi.org/10.1016/j.ijhydene.2025.02.072.
[27] Tatar SM, Akulker H, Sildir H, Aydin E. Optimal design and operation of integrated microgrids under intermittent renewable energy sources coupled with green hydrogen and demand scenarios. Int J Hydrogen Energy Jul. 2022;47(65): 27848–65. https://doi.org/10.1016/j.ijhydene.2022.06.130.
[28] Takatsu N, Farzaneh H. Techno-Economic analysis of a novel hydrogen-based hybrid renewable energy System for both grid-tied and off-grid power supply in Japan: the case of Fukushima prefecture. Appl Sci Jun. 2020;10(12):4061. https://doi.org/10.3390/app10124061.
[29] Gandhi O, Rodriguez-Gallegos CD, Srinivasan D. Review of optimization of power dispatch in renewable energy system. In: 2016 IEEE innovative smart Grid Technologies - asia (ISGT-Asia). IEEE; Nov. 2016. p. 250–7. https://doi.org/10.1109/ISGT-Asia.2016.7796394.
[30] Shao Z, Cao X, Zhai Q, Guan X. Risk-constrained planning of rural-area hydrogen-based microgrid considering multiscale and multi-energy storage systems. Appl Energy Mar. 2023;334:120682. https://doi.org/10.1016/j.apenergy.2023.120682.
[31] Ngila Mulumba A, Farzaneh H. Techno-economic analysis and dynamic power simulation of a hybrid solar-wind-battery-flywheel system for off-grid power supply in remote areas in Kenya. Energy Convers Manag X Apr. 2023;18:100381. https://doi.org/10.1016/j.ecmx.2023.100381.
[32] Hadidian Moghaddam MJ, Kalam A, Nowdeh SA, Ahmadi A, Babanezhad M, Saha S. Optimal sizing and energy management of stand-alone hybrid photovoltaic/wind system based on hydrogen storage considering LOEE and LOLE reliability indices using flower pollination algorithm. Renew Energy May 2019; 135:1412–34. https://doi.org/10.1016/j.renene.2018.09.078.
[33] HassanzadehFard H, Tooryan F, Collins ER, Jin S, Ramezani B. Design and optimum energy management of a hybrid renewable energy system based on efficient various hydrogen production. Int J Hydrogen Energy Nov. 2020;45(55): 30113–28. https://doi.org/10.1016/j.ijhydene.2020.08.040.
[34] Babatunde OM, Munda JL, Hamam Y. Hybridized off-grid fuel cell/wind/solar PV/battery for energy generation in a small household: a multi-criteria perspective. Int J Hydrogen Energy Feb. 2022;47(10):6437–52. https://doi.org/10.1016/j.ijhydene.2021.12.018.
[35] Suresh V, , M. M., Kiranmayi R. Modelling and optimization of an off-grid hybrid renewable energy system for electrification in a rural areas. Energy Rep Nov. 2020; 6:594–604. https://doi.org/10.1016/j.egyr.2020.01.013.
[36] Pan G, Gu W, Qiu H, Lu Y, Zhou S, Wu Z. Bi-level mixed-integer planning for electricity-hydrogen integrated energy system considering levelized cost of hydrogen. Appl Energy Jul. 2020;270:115176. https://doi.org/10.1016/j.apenergy.2020.115176.
[37] Ji M, Zhang W, Xu Y, Liao Q, Jaromír Klemeš J, Wang B. Optimisation of multi-period renewable energy systems with hydrogen and battery energy storage: a P-graph approach. Energy Convers Manag Apr. 2023;281:116826. https://doi.org/10.1016/j.enconman.2023.116826.
[38] Barhoumi EM, Okonkwo PC, Ben Belgacem I, Zghaibeh M, Tlili I. Optimal sizing of photovoltaic systems based green hydrogen refueling stations case study Oman. Int J Hydrogen Energy Sep. 2022;47(75):31964–73. https://doi.org/10.1016/j.ijhydene.2022.07.140.
[39] Knbs. "2019 Kenya Population and Housing Census Volume II: distribution of Population by Administrative Units," [Online]. Available: https://www.knbs.or.ke/2019-kenya-population-and-housing-census-reports/; 2020.
[40] K. N. B. of Statistics. 2019 Kenya Population and Housing Census Summary report on Kenya's population projections. Nairobi, https://www.knbs.or.ke/wp-content/uploads/2023/09/2019-Kenya-population-and-Housing-Census-Summary-Report-on-Kenyas-Population-Projections.pdf; 2023.
[41] Kirimi M, Gitau JK, Mendum R, Muthuri C, Njenga M. Cleaner cooking with charcoal in Kibera informal settlement in Nairobi, Kenya, and its implications for livelihoods and the environment. Energies Sep. 2023;16(19):6808. https://doi.org/10.3390/en16196808.
[42] Agency IE. Electricity generation in Kenya [Online]. Available: https://www.iea.org/countries/kenya/electricity; 2023.
[43] Thronjo P. Nairobi's darkest estate: where tenants use torches during the day. The Standard, Nairobi, https://www.standardmedia.co.ke/article/2001253600/nairobi-s-darkest-estate-where-tenants-use-torches-during-the-day; 2022.
[44] Wambugu P. Kenya Green Hydrogen Strategy & Roadmap. Nairobi: Kenya Electricity Generating Company (KenGen); 2023 [Online]. Available: https://www.ctc-n.org/sites/default/files/Session3-KenyaGreenHydrogenStrategy-KenGen.pdf.
[45] Müller LA, Leonard A, Trotter PA, Hirmer S. Green hydrogen production and use in low- and middle-income countries: a least-cost geospatial modelling approach applied to Kenya. Appl Energy Aug. 2023;343:121219. https://doi.org/10.1016/j.apenergy.2023.121219.
[46] Oyan R, E, Kantel A, Jaroudi I, Müller V, Aboushady N, Altenburg T, Stamm A, Strohmaier. Towards a green H_2 economy: kenya country report. Nairobi: Fraunhofer Institute for Systems and Innovation Research ISI & German Institute of Development and Sustainability (IDOS); 2024. HYPAT Working Paper 13/2024), https://hypat.de/hypat-wAssets/docs/new/publikationen/hypat_country-report_kenya.pdf.
[47] Lubello P, et al. Green hydrogen futures in LMICs: opportunities for fertilizer and steel production in Kenya. iScience Apr. 2025;28(4):112298. https://doi.org/10.1016/j.isci.2025.112298.
[48] S. U. H2Global Foundation, Fraunhofer IEE. Renewable ammonia: kenya's business case. H_2Global White paper, v11; 2025 [Online]. Available: https://h2-global.org/wp-content/uploads/2025/06/H2Global-White-paper-Kenyas-Business-Case-v11.pdf.
[49] H. & F. C. T. O. U.S. Department of Energy. "DOE hydrogen program record 24004: Heavy-duty fuel cell system cost – 2023 (Record No. 24004). Energy Efficiency and Renewable Energy 2024 [PDF], https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/review24/24004-hd-fuel-cell-system-cost-2023.pdf.
[50] Chang X, Xu Y, Sun H, Khan I. A distributed robust optimization approach for the economic dispatch of flexible resources. Int J Electr Power Energy Syst Jan. 2021; 124:106360. https://doi.org/10.1016/j.ijepes.2020.106360.
[51] Alshammari N, Asumadu J. Optimum unit sizing of hybrid renewable energy system utilizing harmony search, Jaya and particle swarm optimization algorithms. Sustain Cities Soc Sep. 2020;60:102255. https://doi.org/10.1016/j.scs.2020.102255.
[52] Yodwong B, Guilbert D, Phattanasak M, Kaewmanee W, Hinaje M, Vitale G. Faraday's efficiency modeling of a proton exchange membrane electrolyzer based on experimental data. Energies Sep. 2020;13(18):4792. https://doi.org/10.3390/en13184792.
[53] Bouguerra A, Badoud AE, Mekhilef S, Kanouni B, Bajaj M, Zaitsev I. Enhancing PEM fuel cell efficiency with flying squirrel search optimization and Cuckoo Search MPPT techniques in dynamically operating environments. Sci Rep Jun. 2024;14(1): 13946. https://doi.org/10.1038/s41598-024-64915-7.
[54] Oyewole OL, Nwulu NI, Okampo EJ. Optimal design of hydrogen-based storage with a hybrid renewable energy system considering economic and environmental uncertainties. Energy Convers Manag Jan. 2024;300:117991. https://doi.org/10.1016/j.enconman.2023.117991.
[55] Wilk M, Magdziarz A. Hydrothermal carbonization, torrefaction and slow pyrolysis of Miscanthus giganteus. Energy Dec. 2017;140:1292–304. https://doi.org/10.1016/j.energy.2017.03.031.
[56] Nguyen NM, Alobaid F, Epple B. Process simulation of steam gasification of torrefied woodchips in a bubbling fluidized bed reactor using aspen plus. Appl Sci Mar. 2021;11(6):2877. https://doi.org/10.3390/app11062877.
[57] Suliman MS, et al. Quantifying the climate Co-Benefits of hybrid renewable power generation in Indonesia: a multi-regional and technological assessment. Climate Feb. 2024;12(2):23. https://doi.org/10.3390/cli12020023.
[58] Rossi CCRS, Cardozo-Filho L, Guirardello R. Gibbs free energy minimization for the calculation of chemical and phase equilibrium using linear programming. Fluid Phase Equilib Apr. 2009;278(1–2):117–28. https://doi.org/10.1016/j.fluid.2009.01.007.
[59] Koukkari P, Pajarre R. Introducing mechanistic kinetics to the Lagrangian Gibbs energy calculation. Comput Chem Eng May 2006;30(6–7):1189–96. https://doi.org/10.1016/j.compchemeng.2006.03.001.
[60] Channiwala SA, Parikh PP. A unified correlation for estimating HHV of solid, liquid and gaseous fuels. Fuel May 2002;81(8):1051–63. https://doi.org/10.1016/S0016-2361(01)00131-4.
[61] Sarkar M, Kumar A, Tumuluru JS, Patil KN, Bellmer DD. Gasification performance of switchgrass pretreated with torrefaction and densification. Appl Energy Aug. 2014;127:194–201. https://doi.org/10.1016/j.apenergy.2014.04.027.
[62] Boutton TW, Tieszen LL, Imbamba SK. Biomass dynamics of grassland vegetation in Kenya. Afr J Ecol Jun. 1988;26(2):89–101. https://doi.org/10.1111/j.1365-2028.1988.tb00960.x.
[63] Aksoy H, Kaptan S. Monitoring of land use/land cover changes using GIS and CA-Markov modeling techniques: a study in Northern Turkey. Environ Monit Assess Aug. 2021;193(8):507. https://doi.org/10.1007/s10661-021-09281-x.
[64] Aldiansyah S, Saputra RA. Comparison of machine learning algorithms for land use and land cover analysis using google earth engine (case study: wanggu watershed). Int J Remote Sens Earth Sci Jan. 2023;19(2):197. https://doi.org/10.30536/j.ijreses.2022.v19.a3803.
[65] Climate.OneBuilding.org. Kenya climate data files – WMO Region 1 Africa. https://climate.onebuilding.org/WMO_Region_1_Africa/KEN_Kenya/index.html. [Accessed 14 April 2024].
[66] Department KM. climate-data-management-Kenya meteorological department [Online]. Available: https://meteo.go.ke/. [Accessed 22 July 2024].
[67] Ibrahim L. Cost benefit analysis of Lake Turkana wind project. Eastern Mediterranean University; 2021 [Online]. Available: https://i-rep.emu.edu.tr/xmlui/handle/11129/6050.
[68] S. B., Ana Pueyo HO. Cost and returns of renewable energy in Sub-Saharan Africa: a comparison of Kenya and Ghana [Online]. Available: https://seors.unfccc.int/applications/seors/attachments/get_attachment?code=LJ4LPK7OW9DZ887FV6IQ0SJ51B907HE4; 2016.
[69] Nasser M, Megahed TF, Ookawara S, Hassan H. Performance evaluation of PV panels/wind turbines hybrid system for green hydrogen generation and storage: energy, exergy, economic, and enviroeconomic. Energy Convers Manag Sep. 2022; 267:115870. https://doi.org/10.1016/j.enconman.2022.115870.

[70] Tyler Stehly DM. Patrick Duffy, hernando. Cost of wind energy review. 2024 Edition 2024 [Online]. Available: https://www.nrel.gov/docs/fy25osti/91775.pdf.
[71] Badgett A, Brauch J, Thatte A, Rachel, Rubin, Skangos. Updated manufactured cost analysis for Proton exchange membrane water electrolyzers [Online]. Available: https://www.nrel.gov/docs/fy24osti/87625.pdf.
[72] Fontalvo VM, Nelson GJ, Pupo-Roncallo O, Sanjuan ME, Gómez HA. A techno-economic assessment for fuel cells hybrid systems in stationary applications. Int J Sustain Energy Dec. 2023;42(1):889–912. https://doi.org/10.1080/14786451.2023.2244605.
[73] Wei M. Reversible fuel cell cost analysis and megawatt PEM cost analysis for H2 grid storage systems [Online]. Available: https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/review22/fc332_wei_2022_p-pdf.pdf?sfvrsn=d1411fd6_0; 2022.
[74] IEA. Global Hydrogen Review: assumptions annex [Online]. Available: https://iea.blob.core.windows.net/assets/2ceb17b8-474f-4154-aab5-4d898f735c17/IEAGHRassumptions_final.pdf; 2021.
[75] IRENA. Renewable energy technologies: cost analysis series [Online]. Available: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2012/RE_Technologies_Cost_Analysis-BIOMASS.pdf.
[76] IRENA. Biomass for power generation [Online]. Available: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2012/RE_Technologies_Cost_Analysis-BIOMASS.pdf; 2012.
[77] Naudé W. South Africa must beware of falling victim to green hydrogen imperialism [Online]. Available: https://www.dailymaverick.co.za/article/2023-12-11-south-africa-must-beware-of-falling-victim-to-green-hydrogen-imperialism/. [Accessed 11 February 2025].
[78] Liu G, et al. Economic analysis of hydrogen energy systems: a global perspective. Heliyon Sep. 2024;10(18):e36219. https://doi.org/10.1016/j.heliyon.2024.e36219.
[79] Khare V, Nema S, Baredar P. Optimization of hydrogen based hybrid renewable energy system using HOMER, BB-BC and GAMBIT. Int J Hydrogen Energy Oct. 2016;41(38):16743–51. https://doi.org/10.1016/j.ijhydene.2016.06.228.
[80] Wang R, Zhang R. Techno-economic analysis and optimization of hybrid energy systems based on hydrogen storage for sustainable energy utilization by a biological-inspired optimization algorithm. J Energy Storage Aug. 2023;66:107469. https://doi.org/10.1016/j.est.2023.107469.